Vocabulary position and trademark lifecycles:
An event-dated corpus and a lead/lag text measure for the commercial economy

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Abstract

All 13.99 million USPTO trademark case files are re-parsed here into event-dated prosecution histories, and every filing scored on how surprising its language is against what its industry filed in the five years before it and in the five years after. The average of the two surprises is the filing's *atypicality*; their difference is its *lead*, positive for wording the industry subsequently moved toward (*leading*) and negative for wording it was already leaving behind (*lagging*).

Within the trademark record, *leading* marks are less likely to remain in commerce. Of the most leading fifth of 3.10 million registrations, 46.7% survive the sworn re-proof of use that a US registration requires between its fifth and sixth year, against 48.1% of the most lagging fifth — a gap of 1.4 percentage points, within Nice class and registration year, with owner-clustered errors and drafting and filer controls ($t = 8.3$). The gap holds across industries and across filer types, but not across time: it was two to three times larger for marks filed in the late 1990s, reversed for those filed 2000–2004, and back from 2008, with the entire swing inside four technology classes and a flat one to four points everywhere else. Within those classes, most of the swing is which themes each era's leading filings were drawn from; inside a theme, the penalty is a steadier two to five points.

0.83% of first-time filers with professional representation ever appear in SEC financial-reporting records, against 0.18% of self-filers. The atypicality of a firm's first registered mark does not predict whether its owner ever reaches SEC reporting once class, year and description length are absorbed; yet both leading ($\mu = 0.39\%$) and lagging ($\mu = 0.44\%$) owners outperform the persistent middle ($\mu = 0.24\%$). Neither axis is a proxy for patenting: across 134,075 matched firm-years, lead correlates with $\log(1 + \text{patents})$ at $r = -0.020$ and atypicality at $r = +0.009$, and within a firm over time at -0.043 and $+0.011$.

Atypicality can be measured on terms or on themes, and the two disagree. Term atypicality predicts both outcomes negatively (-0.09pp of SEC reporting per standard deviation of the score; -0.055 standard deviations of log patenting); theme atypicality does not predict SEC reporting and predicts patenting weakly the other way ($+0.012$ standard deviations). The two therefore disagree in sign on both outcomes, which makes a text measure's sign at the firm level a property of how the text was counted rather than of the firm. Every estimate is conditional on the office having granted the mark, because registration is itself selected on language; without that conditioning the same data support weaker and sometimes opposite conclusions. Code, panels, and the corpus build are public at `https://github.com/ericsilver/tm-vocabulary`, with a browsable explorer for the per-theme and per-Nice-class results behind the pooled estimates.

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1 Introduction

Text-based measurement has reshaped how economists quantify invention. Kelly et al. (2021) score every US patent since 1840 by its textual distance from prior and similarity to subsequent patents, and Hoberg & Phillips (2016) build product-market industries from 10-K text; Kalyani (2024) measures patent creativity as the share of technical terminology not seen before and ties it to firm productivity. Both programs inherit the patent record's coverage: technical invention by the small minority of firms that patent. Most firms never patent, service-sector and business-model innovation rarely patents, and patenting propensity varies so much across industries that cross-industry comparisons are fraught (Dinlersoz et al., 2018; Graham et al., 2013). Trademark filings offer the complementary window. Nearly every commercially serious product or service introduction in the United States is accompanied by a trademark filing, the filing's goods/services description states in plain commercial language what the firm intends to sell, and the corpus extends across every sector of the economy at the rate of hundreds of thousands of granted filings per year.

The trademark–innovation literature is three decades old and has moved well past counting (Castaldi, 2020). Flikkema et al. (2025) survey its first ten years of validation work and its open questions; Willeke et al. (2026) inventory the data sources and the preprocessing burden that has kept them underused. Micro-level links to firm dynamics are established (Dinlersoz et al., 2018, 2023), and the goods/services description is already read as text rather than treated as a label: Semadeni (2006) scores consulting service-mark descriptions into a competitive positioning space, Semadeni & Anderson (2010) tracks whether one firm's described offering is imitated by rivals, and Flikkema et al. (2019) codes descriptions against the innovation each mark accompanies. Prosecution records have been exploited too: Fink et al. (2024) use contested filing sequences to study trademark races, and their outcome — continued use five years after registration — is the margin this paper's central result also runs on. What Flikkema et al. (2025) put as the field's next step is the move from counts and innovation intensity to the *type*, *quality*, and *trend* content of what marks cover: the trademark analogue of what Kelly et al. (2021) and Kalyani (2024) did for patent text. That analogue does not yet exist at corpus scale, and the two strands above — description text and prosecution events — have run on separate data.

This paper builds both on one corpus, so that what a filing says and what subsequently happened to it are observed on the same record.

Whether a description's language is strange for its industry predicts very little; whether it arrived *before* its industry's language moved that way predicts a good deal. Marks whose wording the industry subsequently adopted are abandoned at the five-year use-proof gate more often than marks written in language the industry was already leaving behind. Their owners also reach SEC financial reporting less often than the owners of the most lagging marks, though both tails do better than the middle.

The USPTO distributes its trademark backfile as bulk XML — 83 archives covering 1884 to 2025 — from which the published research files retain a filing's current status but not the dated sequence of events that produced it. Re-parsing the backfile yields 242 million prosecution events — every step in the examination of an application — across 13.99 million case files, each event carrying a code and a date: every office action, response, maintenance filing, and cancellation. The event codes are resolved into a 760-code dictionary. A header pass extracts counsel of record, the statutory filing basis, and the postregistration declarations. A third pass recovers owner domicile, which the class-level files omit. Owner names are then normalized to a clustering key and resolved to two external universes: SEC registrants, through the agency's own company-name and ticker files, and Regulation D issuers, through the Form D quarterly data sets — the latter filtered to operating companies, with ambiguous names dropped rather than guessed. That link is what lets a mark's language be followed to financing and to public listing. Where the patent record dates invention once, at grant, the trademark prosecution record dates a commercial claim's entire life, including the use-proof gates at which marks attached to products that never materialized are culled.

Every filing is grouped by its industry class — one of the 45 international goods and services categories set by the Nice Agreement — and placed by its filing date, then put to two readers who have seen disjoint halves of that industry's stream: one who knows only what the class filed before it, one who knows only what it filed after. Each reader's "how surprising" is a Kullback–Leibler divergence between the filing's mix of topics — its weights over word-clusters fitted to the corpus — and the industry's, large when the filing puts weight where its industry's filings put little. *Atypicality* is the average of the two answers. *Lead*, written ΔKL, is the gap. Language that surprised the first reader and not the second was surprising because it came first: the industry moved toward it. Language that read as ordinary to the first and unusual to the second was left behind as the field moved on. Filings of the first kind are *leading*, of the second *lagging*.

The construct adapts Murdock et al. (2017) and Barron et al. (2018); Section 3 sets out what transfers and what does not. The nearest measurement precedent on this corpus is von Graevenitz et al. (2022), who flag goods/services tokens that are both new and fast-spreading; their unit is the token and their flag binary, while the score here is continuous and attaches to the filing, so every application carries a position that can be crossed with its own prosecution outcomes.

Results are stated on marks the office granted. Registration is itself selected on language, so a correlation over all filings mixes what examination does to a description with what the market does to the product; Appendix E reports the unconditional version. Two further products of the exercise are released with it: the corpus, and a measurement hazard that applies to text-based novelty measures generally.

2 Registration measures follow-through; the use-proof gate measures survival

A US trademark passes through a sequence of dated, high-stakes steps, and each step leaves a record. An application is filed with a goods/services description; an examiner reviews it, often issuing office actions the applicant must answer; if allowed and (for intent-to-use filings) a statement of use is filed, the mark registers. Registration starts a maintenance clock: in years five to six the owner must file a Section 8 declaration proving continued use in commerce, with a six-month grace period; around year ten a combined Section 8 declaration and Section 9 renewal is due, and every ten years thereafter. A mark that misses a use-proof gate is cancelled. Madrid Protocol registrations (§66(a) basis) file Section 71 declarations on the same schedule and are flagged separately throughout. Graham et al. (2013) describe the administrative data; the event-dated corpus built here adds the full prosecution history of each case, so a cancellation is assigned to a specific gate by its date rather than inferred from a status snapshot.

The paper relates two properties of a filing's description — how unusual its language is for its industry, its *atypicality*, and whether that language is ahead of or behind where the industry's language was moving, its *lead* — to what happened to the filing afterwards. Section 3 builds the two properties. The outcomes are three steps in the sequence above, each read from the dated record and each bounded as follows.

Registration. Whether an application reaches registration at all, among filings made 1995–2018. Of the applications that do not register, roughly nine in ten die by the applicant's own inaction — never filing the statement of use, or going silent during prosecution — and only 4.3% are removed

Table 1: The trademark funnel in this corpus. Registration counts are unique serials; gate outcomes are event-dated (a Section 8 cancellation at registration age 4.0–8.5 years). Source: the re-parsed USPTO application backfile, scored as in Section 3.

Stage	n	Share
Debut filings scored, 1995–2018 (owners' first filings)	3,589,068	
reached registration	2,066,600	57.6%
All unregistered class-records filed 1995–2018	3,798,318	
never filed statement of use		42.6%
stopped responding in prosecution		45.3%
removed adversarially		4.3%
Registrations 2002–2018, scored, unique serials	3,104,534	
failed the first use-proof gate (event-dated)		52.3%
Passed first gate, cohorts 2002–2013, terminal status known	1,129,724	
failed at the year-ten renewal		39.1%

adversarially (Table 1), so registration is read as a launch indicator rather than an examiner's verdict.

Survival of the first use-proof gate. Whether a registration was cancelled for non-use at registration age 4.0 to 8.5 years, among registrations issued 2002–2018. This is where the market's verdict reaches the record: half of registered marks do not survive it, and failing it means the product the mark named is no longer in commerce. Every gate estimate is conditional on the mark having been granted, because registration is itself selected on language; what the unconditional composite looks like is shown once, in Appendix E, and not otherwise used.

Reaching SEC reporting. Whether the owner of a first-time filing ever appears in SEC financial-statement records, a firm-level outcome used in Section 4.5 to ask whether either property of the text marks firms bound for public markets.

Outcomes are rates, so every estimate is a shift in a probability and is stated in percentage points against the base rate it shifts: +1.4pp on a 52.3% base means 53.7% against 52.3%, not 52.3% multiplied by 1.014. The basic comparison is the outcome rate of the fifth of filings with the highest lead minus that of the fifth with the lowest, with the fifths cut inside a single industry and year so that no difference between industries or between years enters; the regression tables add drafting and filer controls and cluster errors on the owner. Where a continuous version is given, it is the change in the outcome for a one-standard-deviation change in lead, in percentage points.

Counting differs by table. A filing may claim several Nice classes at once, so a *class-record* counts it once per class while a *registration* counts it once by serial number; a *debut filing* is an owner's first, one per owner. The funnel above counts debut filings, the gate regressions unique registrations, the per-class figures class-records.

Three windows bound the samples, each set by an institutional fact. Scoring runs 1995–2019, because a full five-year past reference does not exist before 1990 and a full forward reference does not exist after 2020; the analysis cut of 1995 is about two years more conservative than the measured burn-in requires (Appendix A). The gate cohorts end in 2018 because the cancellation record runs to April 2026 and the failure window closes at registration age 8.5 years, so 2018 is the last cohort whose window has essentially elapsed — the top of that range is fractionally censored, and a separate 2019 replication is reported below. They begin in 2002 because that is the conventional cut for this corpus, and that lower bound turns out to matter: carried back to the first scoreable cohorts the penalty is two to three times larger than it has ever been since, and Section 4.3 reports what the restricted window concealed.

Outcomes are read from the event-dated prosecution history, which attributes each cancellation to a specific gate and observes counsel and basis directly, and cross-checked against the USPTO status-code dictionary.[1]

3 The construct: how surprising a description is, and when

3.1 Two readers, one living backwards

Imagine two readers of a filing's goods/services description, each having seen a disjoint half of its industry. One has read only what the industry filed in the five years before it. The other, living backwards as Merlin was said to, has read only the five years after. Each is asked how surprising the wording is. The measurement is how much they disagree: a description that startles the first reader and not the second arrived before its industry's language moved that way.

Both windows are anchored on the filing's own date — the 1,826 days before it and the 1,826 days after — so no two filings made on different days share a reference, and filings made on the same day fall in neither of the other's windows. Inside a window every day counts the same, so a term the class adopted five years ago and one it adopted last month are equally familiar. The measure asks whether wording is unusual against a period, not whether it is ahead of a trend inside that period; it has no momentum channel.

3.2 From a description to two numbers

The apparatus is that of Murdock et al. (2017), developed on Darwin's reading notebooks, and Barron et al. (2018), who extended it to French Revolution parliamentary debates. They score a document against those before it and again against those after, calling the two quantities *novelty* and *transience* and their signed difference *resonance*. Their object was the life of ideas: how often an idea appeared, and how ideas stood in relation to one another, across one reading life or one assembly's debate. The object here is the composition of commercial offerings — which components a product or service is described as having, and how that composition shifts across an industry over time. The arithmetic carries over unchanged. The interpretation does not: their document has one author and the reference stream is that author's own reading, so a high value says something about a mind, while here a filing is scored against other firms' filings and its author is usually counsel drafting for scope of protection.

What is being measured, before the arithmetic. The quantity is information content in Shannon's sense. Something that happens with probability p carries $-\log p$ of information when it happens: a common thing carries little, a rare thing a lot. Here the things are the themes a description draws on, and the probabilities are how often the filing's own industry drew on each of them in the reference window. A filing's score is the average information content of its themes under its industry's frequencies, weighted by how heavily it uses each, less the same average taken under the filing's own frequencies. It is zero for a description composed exactly as the industry composes its descriptions and grows as the filing puts weight where the industry puts little. Its purpose is to place each filing against its industry's language at two moments, so that being different (atypicality) can be separated from being early (lead).

[1] 700 registered; 701–705 Section 8 accepted; 710 cancelled for Section 8 failure; 800 renewed; 900 expired.

Counting words. A description is reduced to a bag of words: the text is lowercased, split into tokens of three or more letters, and counted, with adjacent pairs counted as tokens in their own right so that *computer software* registers alongside *computer* and *software*. Order is discarded. Take the filing for AMAZON.COM, lodged in the advertising and retail class on 18 April 1997:

> computerized on line search and ordering service featuring the wholesale and retail distribution of books, music, motion pictures, multimedia products and computer software in the form of printed books, audiocassettes, videocassettes, compact disks, floppy disks, CD ROMs, and direct digital transmission

Of its 45 distinct terms, 33 are in the theme model's vocabulary, among them *computerized*, *search*, *ordering*, *ordering service*, *retail*, *distribution*, *printed books* and *digital transmission*.

Grouping words into themes. Each filing is then re-described in a fixed, much smaller set of coordinates. Latent Dirichlet allocation is run once over a stratified sample of the whole corpus, all 45 classes together: it learns 50 clusters of words that tend to occur together — called *themes* here — and, for any filing, the proportions in which its words draw on them. Nothing supervises this; the clusters come from co-occurrence alone. The themes are shared across classes; what is specific to a class is the reference, the average theme proportions of that class's own filings in the window. The Amazon filing draws on six themes at more than a fiftieth of its weight: 0.41 on media and entertainment goods (*video, computer, electronic, music, games, audio*), 0.15 on computer and information services (*services, computer, software, information, data*), 0.13 on retail store services (*store, retail, store services, featuring*), 0.10 on printed matter (*paper, books, printed, cards, stationery*), 0.09 on downloadable and online software, and 0.07 on marketing of consumer goods. The online appendix lists every theme with its leading words and its weight in each class.[2]

Fifty is coarse, roughly one theme per industry, and the paper does not rest on it. The model is refitted at 100, 200 and 500 themes and the corpus rescored; Appendix B reports what changes. Signs and significance do not. Magnitudes in a single class do, by up to a factor of two, and not monotonically in the theme count, which a second fit at the same count but a different random seed shows to be mostly fitting noise. Fifty is kept throughout because it is the coarsest resolution at which every result holds, so nothing reported depends on a fine partition, and because its lead contrast is the lowest of the four, so it is the conservative choice.

Comparing against the two references. Write P for the filing's distribution over themes, and Q_{past} and Q_{future} for that distribution averaged over every same-class filing in the two windows, $[d_i - W, d_i)$ and $(d_i, d_i + W]$, with $W = 1{,}826$ days. How far P sits from a reference Q is the Kullback–Leibler divergence,

$$\mathrm{KL}(P \,\|\, Q) \;=\; \sum_k P_k \log \frac{P_k}{Q_k},$$

a sum over the 50 themes, zero when the two distributions agree and growing as the filing puts weight where the industry puts little. It is measured in nats, the natural-log unit. Write

$$K^- = \mathrm{KL}(P \,\|\, Q_{\text{past}}), \qquad K^+ = \mathrm{KL}(P \,\|\, Q_{\text{future}}),$$

so K^- is what the first reader feels and K^+ the second: the filing's past-facing and future-facing surprise.

[2] `https://aporia.institute/tm-vocabulary/online-appendix/themes_T50.html`

Taking the difference. Amazon's filing is scored against 41,166 earlier filings and 146,532 later ones, giving $K^- = 1.286$ and $K^+ = 1.163$. The two axes used throughout are their average and their signed difference:

$$\underbrace{A \;=\; \tfrac{1}{2}\left(K^- + K^+\right)}_{\text{atypicality}}, \qquad \underbrace{L \;=\; K^- - K^+}_{\text{lead}}.$$

For Amazon, $A = 1.225$ and $L = +0.123$. Against its own class those say something the raw pair does not: its atypicality is at the 47th percentile, entirely ordinary, while its lead is at the 94th. The industry moved toward this description in the five years after it was filed and had not been writing that way in the five years before. Nothing about the wording was strange; it was early. L is signed, positive for a leading filing and negative for a lagging one, and the symbol ΔKL is used interchangeably with it.

What "early" means is visible in the themes themselves. Of the six Amazon drew on, the share of class-035 filings drawing on each, in the five years before its filing and the five after:

Theme	Amazon's weight	Before (%)	After (%)
Retail store services	0.13	42.6	50.3
Computer and information services	0.15	58.9	59.9
Media and entertainment goods	0.41	9.9	11.3
Downloadable and online software	0.09	2.3	4.1
Printed matter	0.10	8.3	7.8
Marketing of consumer goods	0.07	10.5	9.3

A theme counts as drawn on when it carries at least a tenth of a filing's weight. The three themes that carry most of Amazon's description — retail, media goods, and the still-small online-software theme — all became more common in the class over the following five years, the last nearly doubling; the two minor themes it also touched became slightly less so. The filing is closer to the class's future composition than to its past one, and that, not any rare word, is its lead.

Differencing is what makes that separation possible. K^- and K^+ correlate at 0.988 on the software class, so entering both is close to entering one variable twice; rotating them into an average and a difference isolates the large common component from the small residual that carries the timing, and leaves the two regressors nearly uncorrelated (+0.036 against each other, where K^- alone would give +0.114). The cost is that L is a small residual of two large quantities, about a sixth of either level's spread, which Appendix A quantifies as a genuine fragility.

Table 2: The quantities, their sources, and the terms used here. *Lagging* and *leading* describe the two signs of L; *atypicality* is unsigned and says nothing about direction.

Quantity	Definition	Murdock et al.	Term used here
Surprise against the past	K^-	novelty	past-facing surprise
Surprise against the future	K^+	transience	future-facing surprise
Their average	$A = \frac{1}{2}(K^- + K^+)$	—	**atypicality**
Their signed difference	$L = K^- - K^+$	resonance	**lead** (leading / lagging)
Its magnitude	$\lvert L \rvert$	—	lead magnitude

Why the words are not scored directly. The themes are an extra step, and a reader may wonder why the words themselves are not compared to the reference: build the filing's distribution over the class vocabulary from the words it uses, do the same for every filing in the window, and take the divergence. The paper reports that scoring as a cross-check. It cannot be the primary one, because a divergence measured on words mostly counts the length of the filing. The divergence is an average surprise less a spread: for any two distributions,

$$\mathrm{KL}(P \,\|\, Q) \;=\; \underbrace{-\textstyle\sum_k P_k \log Q_k}_{\text{how unexpected these words are}} \;-\; \underbrace{\left(-\textstyle\sum_k P_k \log P_k\right)}_{\text{how spread out the filing is}} \;,$$

and the spread of a filing over words is bounded by how many distinct words it has: a description using k terms cannot spread over more than k of them, so its spread cannot exceed $\log k$, while the reference is spread over about 900 terms. A five-word description is charged the whole difference in width — roughly five nats — before any of its content is considered. Measured rather than derived: take 3,000 software-class filings of at least 150 words, score each against one fixed reference, then truncate the same filings to a random handful of words and score them again. Across a fiftyfold cut in length the surprise term moves by 0.016 nats and the spread term by 2.9. Synthetic filings whose words are drawn at random *from the reference itself* — maximally ordinary by construction — score 5.72 at three tokens and 1.65 at four hundred, tracking log(distinct terms) to within a fiftieth of a nat, and the average real filing scores marginally *below* that null at its own length. Term-scored atypicality correlates with log description length at -0.65 in the software class.

Themes break the coupling because the 50 coordinates are always occupied. Under term scoring the effective number of coordinates a filing spreads over rises seventeenfold from the shortest descriptions to the longest; under themes it varies by less than a third, and not even monotonically — a three-word fragment spreads across *more* themes than a long description, since the model cannot tell which one it belongs to. The width term therefore stops carrying length, and what is left is the part that was always about content. Theme-scored atypicality correlates with log length at -0.08 in the software class and $+0.06$ in advertising and retail, against -0.65 and -0.59 under terms.

What the themes see, and what they do not. The score compares a filing's theme proportions to the average proportions of its industry, so it reads which themes a filing draws on and how heavily. It does not, in any material way, read how a filing *combines* them. Taking each filing's two leading themes and asking how often that pair occurs together relative to the two themes' separate frequencies: how rare the themes are individually explains 70% of the variance in atypicality, and the unusualness of the pairing adds under two points on top of it. What little it adds runs the wrong way for a combination story — pairs that co-occur *more* than chance score higher, which is concentration rather than novelty of arrangement. A filing assembled from ordinary parts in a genuinely new configuration is therefore close to invisible to this measure, which is what Appendix G shows for the Amazon filing one case at a time.

The combination can, however, be scored in its own right, and it is worth knowing what that returns. For a while it was fashionable for a venture to describe itself as "Uber for" something — two familiar things in a pairing that was rare when it was announced and common soon after. The same two windows can be asked about a pairing: take a filing's two heaviest themes and measure how much more or less often they occur together than their separate frequencies would predict, in the class's filings of the five years before and the five after. The negative log of that ratio is the pairing's information content, and its past-minus-future difference is a *pair lead*, positive when the class moved toward the pairing. In the software class, filings in the top fifth of pair lead fail the first use-proof gate 1.7 points more often than those in the bottom fifth (SE 0.2), against 3.2 points for lead itself; the two are nearly uncorrelated ($r = -0.16$), and the pairing's penalty is essentially unchanged (1.6 points) once the filing's own lead and atypicality are taken out. Being early with an arrangement is penalised in its own right, at about half the rate of being early with the parts, and the two are separate pieces of information. The pairing's *rarity*, by contrast, adds nothing once the parts' atypicality is held (0.4 points, $t = 1.9$). The main text uses lead and atypicality of the parts throughout; the pair measure is reported here as a bound on what the construct leaves on the table, not as a third axis.

3.3 Within class, not across it

Goods/services descriptions are drafted to secure scope of protection, not to describe a product distinctively, and counsel reuse language — from the USPTO's own Acceptable Identification of Goods and Services Manual, from house precedent, and from competitors' successful filings. Identical wording across unrelated firms is therefore common, and it carries through to the scores: 17% of scored registrations share a value with another filing in the same class and cohort, which is why the tie rule is stated and tested (Section 4.2). Within a class, the measure picks up departure from a shared drafting convention. Across classes it does not: a class where the Manual supplies dense standard language shows a compressed distribution of atypicality, and one where firms describe novel services in prose a wide one, for reasons unrelated to how innovative either industry is. Every estimate is therefore taken within class and year, and magnitudes compare across classes only as signs and orderings.

The vocabulary is positional rather than causal. A leading filing was ahead of its industry's language, which implies a trajectory without asserting the filing caused it. *Atypicality* is likewise textual, and is not trademark law's distinctiveness doctrine, which governs registrability along the generic-to-fanciful spectrum; the two attach to different objects, one to the mark and one to the description of the goods it covers.

4 Among granted marks, being early costs and being unusual does not pay

4.1 Why these results condition on grant

Registration is not a neutral filter on language, so every estimate below conditions on it. Completion is an inverse-U in atypicality: applicants whose language sits far from their category in either direction follow through least often, and the penalty is larger for the conventional tail than the unusual one. Pooling granted and abandoned filings therefore mixes what examination does to a description with what the market does to the product, and the resulting gradients are attenuated and in places reversed (Appendix E).

Conditioning would be the wrong move if abandonment were usually a pause rather than an ending — if firms reworked refused applications and refiled them, the same commercial intent would be counted once as a failure and again as a success. It is not. Among 2.28 million abandoned applications, only 3.6% are followed within six years by the same owner filing the same mark again in the same class, and only 1.5% by one that then registers. Crediting every successful refiling as an eventual registration moves the inverse-U in atypicality from a depth of +2.91pp to +2.85pp. Abandonment is overwhelmingly terminal, and what the registration step selects on is therefore a property of the filing rather than an artifact of counting.

What examination rewards is a middling specificity — concrete enough to survive a definiteness objection, conventional enough to avoid a likelihood-of-confusion refusal. Completion runs from 46.6% at the least atypical end to about 58% in the middle and back to 53.4% at the most atypical;

Appendix D gives the profile on each axis and why familiarity alone cannot produce it. The rest of this section is about what happens to marks after the office has granted them.

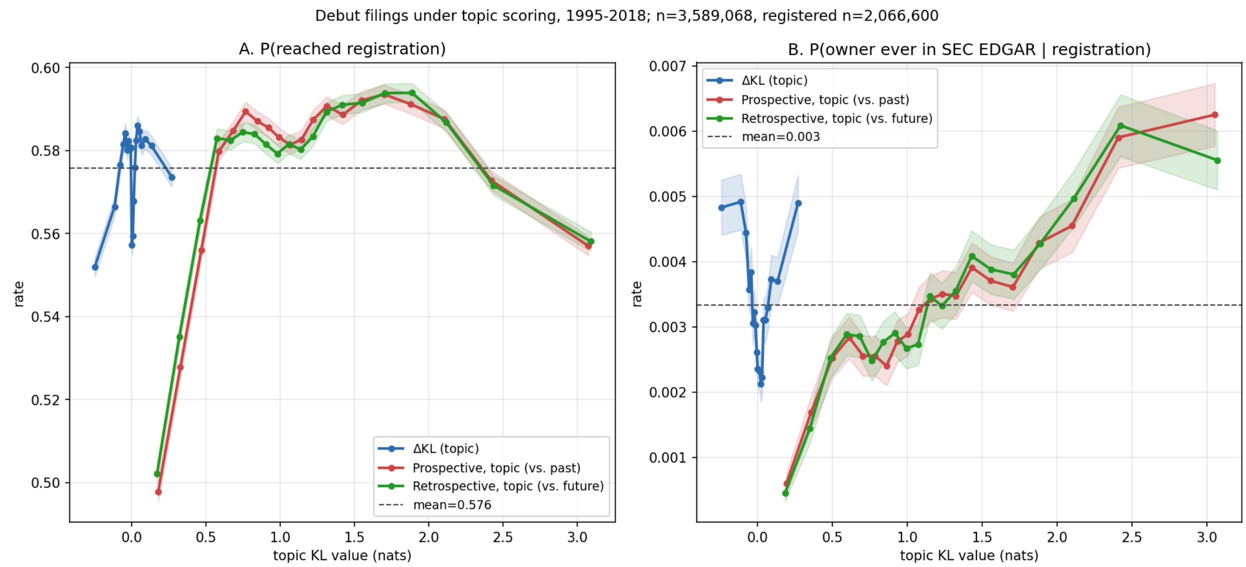

Figure 1: Outcomes for debut filings (owner's first-ever filing) across all 45 Nice classes under topic scoring, filing years 1995–2018, in quantile bins with 95% intervals. $n = 3{,}589{,}068$; registered subset $n = 2{,}066{,}600$. *A:* P(reached registration). *B:* P(owner ever in SEC EDGAR, the Commission's public filing system | registration); the structure in panel B is decomposed in Section 4.5.

4.2 Leading marks fail the use-proof gate

Marks written in language their industry subsequently moved toward fail their first use-proof gate more often than marks written in language it was moving away from. Within Nice class and registration year, controlling for whether the filer drafted the mark themselves or hired counsel, and for the length of the description, marks whose language most leads their industry fail 1.4 points more often (53.3%) than those in the most lagging fifth (51.9%; $t = 8.3$, Table 3).

Among 4.12 million registration class-records 2002–2018, dated first-gate cancellation rises from 50.0% at the bottom topic-ΔKL quintile to 52.4% at the top, positive in 33 of 44 classes with sufficient volume (Figure 3). But a raw pooled contrast mixes the within-industry gradient with class and registration-year composition, treats multi-class registrations as independent observations, and ignores that gate outcomes correlate within owner. The primary estimates therefore come from a linear probability model on 3.10 million unique registrations, one row per serial, with ΔKL quintiles cut *within* Nice class and registration year. The specification adds class×registration-year fixed effects; controls for counsel of record, intent-to-use basis, log description length, log owner filing volume, owner domicile, and foreign filing basis;[3] and standard errors clustered on 1.27 million normalized owners (Table 3).[4]

[3] A US application is filed either on *use* — the mark is already in commerce — or on *intent to use*, which defers proof until after allowance. Foreign applicants may instead file on a home registration (§44(e)) or extend an international registration into the US under the Madrid Protocol (§66(a)); the latter maintain under §71 rather than §8. These are statutory routes to the same registration, and they differ sharply in who files and how often the mark is abandoned,

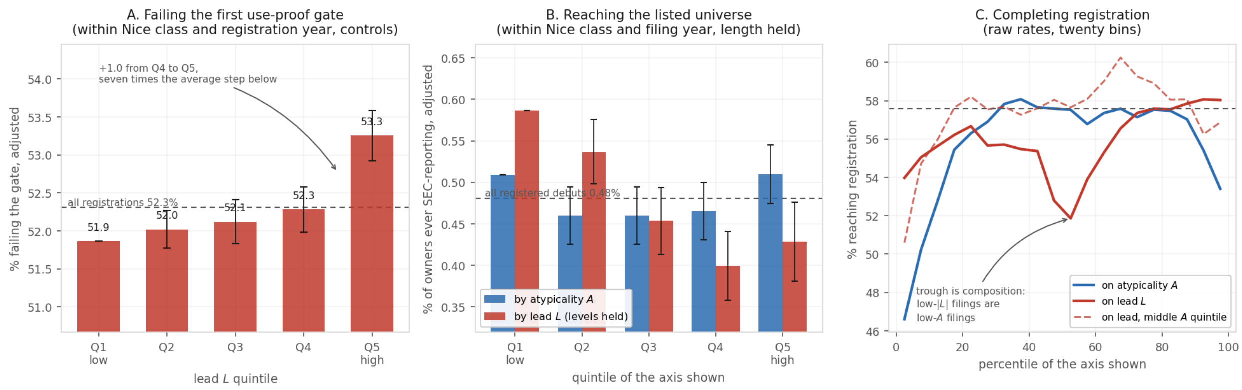

Figure 2: Quintile profiles for the three outcomes. *A:* the share of registrations failing the first use-proof gate, by lead quintile, adjusted within Nice class and registration year with the drafting and filer controls of Table 3; the dashed line is the rate for all registrations. The profile rises gently and monotonically through the fourth quintile and then steps up: about seven tenths of the top-to-bottom gap is the single step from the fourth quintile to the fifth, seven times the average step below it. A gradient is present, but the step at the top dominates it, and the text reports top-versus-bottom contrasts for that reason. *B:* the share of first-time filers whose owner ever reaches SEC reporting, by quintile of atypicality and of lead, adjusted within Nice class and filing year with description length held; flat in atypicality, and falling in lead through the fourth quintile once the levels are controlled. Whiskers in A and B are 95% intervals on the difference from the bottom quintile. *C:* raw registration rates for debut filings in twenty bins — an inverse-U on atypicality, and on lead a trough just above the median that is compositional, absent once atypicality is held in its middle fifth (dashed). Horizontal line at the pooled rate.

The gradient is monotone within cells. Whether the filing was drafted by counsel or by the owner is held because it moves both sides of the comparison at once: self-filers fail the gate at 69.6% against 46.8% for represented owners, and they write shorter descriptions that sit closer to their class's usual language yet lean slightly further ahead of it, so without the control, part of any lead gradient is a self-filing gradient. With it, the penalty is of the same order in both strata (+1.8pp self-filed, +1.2pp counselled). The full control set cuts the magnitude by about a third without absorbing it, and the estimate is if anything *largest* in the cleanest stratum — counsel-represented, US-domiciled owners — so it is not carried by the self-filed or foreign mass-filing segments whose gate failure rates are high for their own reasons.[5] About forty percent of the raw pooled contrast is composition: +1.4pp design-based against +2.5pp raw.

Within the 2002–2018 window the penalty looks modern: indistinguishable from zero for the 2002–2005 cohorts, turning positive in 2006, and running +1.8 to +5.6pp for every cohort from

which is why each enters as a control.

[4]Ties are common. Within the class×registration-year cell the quintiles are cut in, 17.1% of scored registrations share a ΔKL value with at least one other filing, and the largest tie group runs to 53. Anchoring the reference window on the filing date does not remove them, because filings that share a class, a filing date and a goods/services description — an owner registering a family of marks at once, on Manual language — have identical distributions scored against identical references. Cut points are fixed by sorting on the score and then on the serial number, which is reproducible but systematic: serials run in filing order. It makes no difference. Sorting the ties the opposite way moves the top-versus-bottom contrast by 0.001pp, and five pseudorandom orderings move it by at most 0.002pp, against a contrast of 2.287pp — a tenth of one percent of the estimate.

[5]The level effects on those covariates are large in their own right: counsel −19.5pp and Madrid §66(a) basis +10.3pp.

Table 3: First-gate failure and ΔKL: design-based estimates. LPM of event-dated first-gate failure on within-class×registration-year ΔKL quintiles (Q1 omitted), unique serials, registrations 2002–2018. Failure is a dated cancellation for non-use at registration age 4.0–8.5 years, under §8 or, for Madrid §66(a) registrations, §71. All specifications include class×registration-year fixed effects; controls are counsel of record, intent-to-use basis, log goods/services length, log owner filing count, owner domicile (US, CN), and foreign basis (§44(e), §66(a)). Base failure rate 52.3% on this sample.

Sample	Specification	Q5−Q1 (pp)	SE	n
All	FE only	+2.29	0.16	3,104,534
All	FE + controls	+1.39	0.17	3,104,534
Counsel-represented	FE + controls	+1.15	0.20	2,414,318
Counsel + US-domiciled	FE + controls	+1.60	0.22	1,888,352
Self-filed	FE + controls	+1.80	0.26	690,216
Excluding Madrid §66(a)	FE + controls	+1.82	0.18	2,924,427
Counsel + US, 2002–07	FE + controls	−0.25	0.33	551,989
Counsel + US, 2008–12	FE + controls	+2.00	0.42	551,924
Counsel + US, 2013–18	FE + controls	+2.65	0.25	784,439
All	continuous, pp per sd of lead	+0.58	0.05	3,104,534
Counsel + US	continuous, pp per sd of lead	+0.63	0.06	1,888,352

Standard errors replace significance stars: with n in the millions every coefficient has $p < 0.01$, so stars would mark each row identically. Errors are clustered on *normalized owner* — names uppercased, stripped of punctuation and of twenty legal suffixes, so ACME CORP. and ACME CORPORATION are one cluster. A single owner files many marks, drafts them in a house style, and abandons them together when the business fails; the within-owner correlation of gate survival is 0.42, so the 1.27 million owner clusters, not the filings, carry the information. The Madrid row drops §66(a) registrations rather than absorbing them in a control, since they maintain under a different statute; the estimate is larger without them.

2008 onward (Figure 4). That reading does not survive carrying the series back: the penalty was two to three times larger for the earliest scoreable cohorts than for any since, which makes the 2002–2005 quiet a trough between two episodes rather than a starting point (Section 4.3).

The gate is close to a fixed appointment rather than a hazard spread over years: of the 1.35 million cancellations recorded for fully observed cohorts, 17 post before registration age 6.5, 96.1% post between 6.5 and 7.0, and the rest after seven. Since the cancellation record ends on 2 April 2026, a cohort is observable only once it has reached seven years, which makes 2019 the last one with usable gate data and leaves 2020 and later with no observed gate at all. Measured on the 6.5–7.0 band, fully observed for every cohort it covers, the penalty runs between +2.1 and +5.6pp with no trend across the decade and overlapping intervals at its two highest points. The 2019 reading rests on the 78,408 registrations issued early enough in that year to reach seven years inside the record, so it is a first-quarter subsample rather than a cohort.

The outcome is an indicator on a window, so a mark that fails outside it counts as a survivor. If leading marks failed at the same rate as lagging ones but sooner, that alone would put them inside the window more often and the measure would report a difference in rates where there was only a difference in timing. It is not. On the 3.15 million registrations whose ninth anniversary is inside the record — for which failure is settled and no window is needed — the contrast is +2.71pp within class and registration year, against +2.65pp on the paper's 4.0–8.5 window and +2.65pp on the narrow band. Conditional on failing, the probability of failing late carries a contrast of −0.01pp ($t = -0.2$) once class and registration year are held. The pooled version of that contrast is +0.27pp and is composition.

The filer population changed over the same period. Within counsel-represented US-domiciled owners, and with the full control set, the earliest cohort third is slightly negative (−0.25pp, SE 0.33) while the two later thirds are clearly positive (+2.00 and +2.65pp, Table 3). The gradient is therefore not an artefact of the post-2015 surge in self-filed and foreign applications.

4.3 The penalty is not modern, and it is not general

Scores exist from 1995 on, which makes registration cohorts from 1996 evaluable, and their gate outcomes have been settled for two decades. Carried back, the contrast runs +6.9pp for the 1996 cohort and rises to +10.1pp for 1999 — two to three times anything observed since — then collapses. The 2002–2005 cohorts the paper had been reading as a quiet baseline are a trough between two episodes, and one of them is the larger.

Scores begin in 1995, so a 1996-cohort registration must have issued within a year of filing; early cohorts are selected on prosecution speed, and speed plausibly travels with drafting in standard language. Holding pendency in a common one-to-three-year band leaves the series unchanged. Indexing by filing year instead, so a cohort contains every prosecution speed, gives +7.9 to +9.6pp for filings 1995–1999 and −0.95pp for filings made in 2000 — a ten-point swing in a single filing year.

Splitting the four classes that carry most technology filings — 009, 035, 038 and 042 — against the other forty-one separates a large moving component from a small steady one (Figure 5):

Filing years	Technology classes	All others
1995–1999	+14.18 (0.36)	+4.48 (0.27)
2000–2004	−1.37 (0.35)	+1.21 (0.24)
2005–2007	+2.04 (0.43)	+1.99 (0.28)
2008–2014	+9.02 (0.25)	+1.32 (0.17)

Outside technology the penalty is a flat +1 to +4pp in every era, including the one in which the pooled series reads as zero. Inside technology it swings by fifteen points: strongly positive for filings made into the late 1990s, negative for those made in the five years after, and positive again from 2008. The pooled null of 2002–2005 is a technology reversal partly offsetting a non-technology positive, not an absence.

What this design cannot say is which cycle the swing belongs to. Marks filed 1995–1999 reach their gate in 2001–2007 and marks filed 2000–2004 reach theirs in 2006–2013, so a story in which unusual language is punished because it was written during an investment boom and a story in which it is punished because the gate falls during a bust both fit these numbers.

4.4 Three of the four classes reverse, and the leading language changes theme

The four-class split pools industries that do not move together. Taking them one at a time, and then assigning quintiles inside progressively finer cells (Table 4):
Electrical and software goods (009) and telecommunications (038) follow the pooled pattern. Business services (035), the class that holds online retail, reverses hardest: −9.7pp for filings made 2000–2004 against +4.6pp in the five years before. Scientific and computer services (042) never reverses. Its contrast is +6.7pp in the era in which the other three are negative and between +8 and +17pp in every other, and by registration cohort it never falls below +3pp (Figure 6). The reversal is an event in goods and in business services; the class that sells technology as a service sat it out.

Table 4: The leading gate penalty inside technology, by class and by cell. Top-minus-bottom lead-quintile contrast in first-gate failure (pp, SE in parentheses), settled registrations, by filing era. The upper block assigns quintiles within each class and era. The lower block pools the four classes and assigns quintiles within the era (raw, as in the text table above), within class×registration-cohort cells, and within theme×class×registration-year cells, where a filing's theme is the one its description loads most heavily on under the production $T = 50$ model. $n = 1{,}248{,}591$.

Class	1995-1999	2000-2004	2005-2007	2008-2014
009 Electrical, software	+18.04 (0.52)	−5.32 (0.51)	+0.25 (0.62)	+11.67 (0.37)
035 Business services	+4.59 (0.69)	−9.70 (0.56)	+2.02 (0.63)	+1.66 (0.36)
038 Telecommunications	+9.91 (1.40)	−4.24 (1.29)	+2.60 (1.48)	+3.42 (0.94)
042 Scientific, computer services	+17.11 (0.57)	+6.70 (0.62)	+8.42 (0.87)	+12.63 (0.47)
Four classes pooled, raw	+14.49 (0.33)	−1.81 (0.31)	+2.56 (0.38)	+9.16 (0.22)
within class×cohort	+14.27 (0.33)	−3.63 (0.31)	+2.73 (0.38)	+7.24 (0.22)
within theme×class×cohort	+4.78 (0.33)	−1.26 (0.31)	+2.56 (0.38)	+4.51 (0.22)

Assigning quintiles within class and registration year rather than within the era pool changes little. Assigning them within theme, class and registration year changes a great deal: the late-1990s contrast falls from +14.5 to +4.8pp, the post-2008 contrast from +9.2 to +4.5pp, and the 2000–2004 reversal from −3.6 to −1.3pp. Two thirds of the first and half of the second are between themes rather than within them. The leading fifth of each era was drawn from particular themes, and those themes carried their own failure rates; the penalty for being leading *inside* a theme is a steadier +2.5 to +5pp, the same order as the penalty outside technology.

Which themes is a matter of record (Table 5). In 1995–1999 the computer, software, information and data services theme held 38% of technology filings and 75% of the leading fifth, and its registrations failed at 58.9% against 46–53% in the other large themes. That one theme is most of the late-1990s penalty. In 2000–2004 it held 12% of the leading fifth. The leading language had moved to video, music and games on electronic media (31% of the leading fifth against 12% of filings) and the internet-services language had become lagging — wording the class was leaving — and inside several themes the lagging fifth now failed more than the leading one: −3.6pp in electronic media, −5.5 in retail store services, −6.9 in apparatus and control systems. From 2008 the leading fifth over-represents downloadable and online software by four to one (15% against 4%), the vocabulary of the application store; those registrations fail at 67.2%, the highest base rate of any theme, and carry the largest within-theme contrast anywhere in the table, +8.4pp. The computer-services theme's own contrast, meanwhile, is positive in every era (+3.6 to +5.9pp) and never reverses.

So the swing has two layers. Across the whole period, a leading filing is a few points more likely to fail than a lagging one in the same theme, class and cohort, in technology as elsewhere. On top of that, each era's leading vocabulary was concentrated in one or two themes whose marks failed at their own rate: internet services in the 1990s, application software after 2008. In the trough between, the language that had been leading became lagging, and the marks still written in it failed with it. The design-based estimates in Table 3 hold class and cohort fixed but not theme, so they carry both layers; a reader who wants only the within-theme penalty should take the lower block of Table 4 as the guide to its size.

Levels and gradients separate cleanly across filer strata. Self-filers fail the gate far more often than represented owners (69.6% versus 46.8%), and intent-to-use registrations more often than

Table 5: The leading gate penalty by theme within the four technology classes. Themes with at least 20,000 settled registrations, labelled by their five highest-probability terms; quintiles assigned within theme and era. Cells with fewer than 3,000 registrations are blank.

Theme (top words)	n	1995-1999	2000-2004	2005-2007	2008-2014
services, computer, software, information, data	451,253	+3.61 (0.52)	+3.56 (0.49)	+4.18 (0.64)	+5.92 (0.37)
video, computer, electronic, music, games	162,290	+2.53 (0.97)	−3.62 (0.89)	−2.85 (1.01)	+3.36 (0.60)
services, featuring, store, retail, store services	131,107	+8.39 (1.09)	−5.49 (1.01)	+2.92 (1.15)	+0.98 (0.66)
services, field, training, estate, educational	95,602	+7.82 (1.09)	+0.99 (1.12)	+6.02 (1.43)	+2.05 (0.83)
apparatus, systems, devices, electronic, control	84,121	+4.80 (1.27)	−6.92 (1.19)	−0.58 (1.41)	+3.65 (0.88)
design, field, systems, development, services	32,517	+10.40 (2.03)	+4.67 (1.82)	+1.50 (2.27)	−2.27 (1.45)
software, online, downloadable, non-downloadable, virtual	29,472	—	—	—	+3.30 (1.05)
bags, clothing, shirts, jackets, leather	28,421	+1.28 (2.59)	−1.55 (2.39)	−4.66 (2.40)	+6.06 (1.36)
research, medical, scientific, testing, purposes	25,897	+12.65 (2.66)	+12.09 (2.03)	−3.82 (2.59)	+5.63 (1.51)
services, health, field, care, information	22,880	+4.94 (2.04)	+1.39 (2.36)	—	+3.50 (1.72)

use-based ones (53.1% versus 50.2%), but the ΔKL gradient itself is similar across these strata once class, cohort, and drafting covariates are held fixed (+1.8pp self versus +1.2pp counseled in Table 3). Per class, the largest penalties sit in tobacco (+26pp, the vaping era), hand tools (+10pp), processed foods (+10pp) and medical apparatus (+9pp), with negative lifts in beverages, transport and construction.

The risk persists at the year-ten renewal, and on the design-based estimates does not attenuate: the lead coefficient is −0.49pp at the first gate and −0.63pp at the second (Table 7). What attenuates is the raw single-class contrast. In the software diagnostic class, where the event-dated first-gate lift on the 2010–2013 cohorts reaches +12.1pp, the conditional lift at the second gate falls to +7.6pp. Pooled across classes the second-gate profile is not a slope at all. Fitted over 872,079 first-gate survivors with a resolved year-ten status, a straight line through renewal against lead explains 1% of the variation across forty bins and a V explains 89% (ΔAIC = 83, the largest margin anywhere in this paper). Renewal is highest at both ends of the lead axis and lowest just below its centre. The −0.63pp linear coefficient reported for that row is therefore close to meaningless as a description, though it is not wrong about the direction of the leading tail. By year ten the lagging tail has closed the gap in cumulative failure, having simply taken longer to get there. Most of the difference resolves at first proof of use.

A competing reading is that the gate measures disengagement rather than products: an owner who has lost interest both answers the examiner slowly and lets the registration lapse. The prosecution record tests this through the gap between the first non-final office action and the response to it.

Response speed does not predict the gate. Across the 1.22 million scored registrations with a usable office-action/response pair, gate passage varies between 46.1% and 48.1% across response-speed quintiles, from a median of seven days to 183 — flat, and if anything the quickest responders pass slightly *less*. The lead penalty is present in both halves: −3.48pp (±0.40) among faster-than-median responders and −2.36pp (±0.40) among slower, against −2.94pp pooled. The restriction drops 61% of the sample, disproportionately applications that never drew a refusal, so this speaks to the prosecuted subset.

Mixing the past and future reference windows identifies which kind of novelty the gate punishes. Scored against a long past and a short future, so that lead measures a break with what the class had established, the top-versus-bottom contrast is +3.0pp; scored against a short past and a long future, so that it measures resemblance to where the class was heading, it is +1.65pp. The symmetric five-year measure sits between, and the penalty rises monotonically along the mix. The gate punishes breaking with the past more than it punishes anticipating the future, but it punishes both (Appendix A).

4.5 Being early costs at the public-reporting margin; being unusual does not pay there

The outcome here is whether an owner ever appears in the Securities and Exchange Commission's financial reporting records. That is broader than exchange-listed: roughly half the matched firms carry a ticker and the rest report for other reasons. The rate rises monotonically in the KL *levels* on both axes, from 0.19% at the bottom past-facing quintile to 0.52% at the top. In signed ΔKL it is U-shaped.

Table 6: Debut listing and vocabulary position: design-based estimates. LPM of P(owner ever in SEC EDGAR) on 1,556,968 registered debut filings 1995–2018, one observation per owner, class×filing-year fixed effects, heteroskedasticity-robust SEs. Quintiles cut within class×year. Base rate 0.481%.

	Coefficient (pp)	SE
Atypicality quintiles, $A = \frac{1}{2}(K^- + K^+)$ *(Q1 omitted)*		
Q5, FE only	+0.017	0.018
Q5, FE + log length	+0.000	0.018
Q5, FE + log length, on K^- alone	−0.003	0.018
Signed ΔKL quintiles, FE + log length + KL levels		
Q3	−0.133	0.020
Q4	−0.187	0.021
Q5	−0.158	0.024
Decomposition, FE + log length		
|ΔKL| (pp per sd)	+0.067	0.009
signed ΔKL (pp per sd)	−0.047	0.006
log description length	+0.056	0.004

Standard errors in place of significance stars; see the note to Table 3.

The raw atypicality gradient does not survive the design-based estimates (Table 6). Within class and filing year, holding description length fixed, top-quintile atypicality raises the probability of SEC reporting by +0.000pp on a 0.481% base ($t = 0.0$), and by −0.003pp on the past level alone. The quintile profile is not even monotone: the three middle quintiles sit *below* the bottom one and the top merely returns to it. The 2.7-fold gradient in the raw bins is composition — firms that go on to report cluster in particular classes, years, and multi-class filings — and class×year fixed effects absorb it. Lead costs, at equal atypicality: the signed component enters negatively (−0.047pp per standard deviation, $t = -7.8$) while the unsigned component enters positively (+0.067pp per standard deviation, $t = 7.7$). The U-shape in raw bins is exactly this pair: lagging filings show the atypicality premium with no lead penalty, leading filings show it net of that penalty, and the middle quintiles show neither. Once the levels are controlled, the apparent recovery of the far-leading tail is much reduced: the quintile profile falls through Q4 and recovers only slightly at Q5 (Table 6 middle panel).

Owners are matched to SEC registrants by normalized exact name, over a candidate universe of all 5.67 million owners in the corpus; names resolving to more than one registrant are dropped rather than assigned, and one match is kept per owner string. That yields 19,889 matched owners covering 7,588 registrants, and a base rate of 0.481% among registered debut filers. Validation on 22 companies that indisputably went public — across software, pharmaceuticals, food, apparel, aerospace and autos — matches all 22. A random sample of 25 matches carrying at least three filings was inspected by hand and none was wrong, which is a check on gross failure modes rather than a precision estimate.

The linkage is nonetheless a name match, and that bounds the atypicality reading in particular. Firms with distinctive names are easier to match than firms with generic ones, and distinctiveness of a name plausibly travels with distinctiveness of the goods/services text; nothing in this design separates the two. Restricting the candidate universe — to a subset of classes, say — makes matchability itself a function of the regressor, and the atypicality gradient is sensitive to exactly that. Its absence here is therefore the more credible reading of the two, and the signed axis, which does not move when the linkage is varied, carries the weight. The financing rows of Table 7 rest on a Form D match whose raw extracts are not in the current tree and should be read as provisional.

In raw rates, 0.444% of the most lagging fifth of registered debut owners later appear in SEC reporting records, against 0.240% in the middle fifth and 0.390% of the most leading fifth: both tails beat the middle, and the lagging tail beats the leading one. Under the design-based specification the profile declines from the lagging end through the fourth quintile and recovers slightly at the leading end, so the ordering is monotone over four of the five quintiles.

The profile is a U rather than a slope, so a linear summary of it points downward through a curve whose tails are its high points; Appendix B fits it: a V with its vertex about a third of a standard deviation above mean lead, explaining 59% of the binned variation against 13% for a straight line. And "SEC reporting" is not "went public": splitting matched owners on whether their SEC filer number carries a ticker gives 10,556 listed against 9,333 reporting without one, and both halves show the same U, so the result is not an artifact of counting non-operating registrants as successes.

Atypicality predicts survival at the use-proof gate but not which owners become SEC-reporting once composition is absorbed. Lead is the timing bet, and it costs at both margins.

The two are not independent in the obvious economic sense. What makes a filing leading is precisely that its industry moved toward its language, and an industry that has moved toward your language now describes its goods the way you describe yours. Being early therefore consumes the quantity that pays: the mark that was unusual in 2011 is ordinary by 2016 not because its owner changed anything but because the category arrived. On this reading atypicality is a position competitors erode, and lead measures how fast. The design cannot demonstrate that mechanism, since it cannot separate a category catching up with a filing from some third thing moving both. What can be said is weaker: a negative coefficient on lead at equal atypicality is what the mechanism would produce if it were real.

4.6 The construct is not patenting

Within the firms that do both, trademark vocabulary and patenting move independently. On a panel of trademark owners name-matched to PatentsView patent assignees (PatentsView, 2024), each axis is regressed on $\log(1 + \text{patents})$ with firms absorbed and errors clustered on the firm, over 134,075 matched firm-years across 17,506 firms. The outcome is standardized, so β is the shift in standard deviations of the axis per unit of log-patents; one unit is roughly a tripling of the patent count, and a firm going from none to twenty moves about three units.

Text counted as	Axis	β (sd)	t	Correlation r
Themes	atypicality	+0.012	+3.0	+0.009
Themes	lead	−0.058	−9.4	−0.020
Terms	atypicality	−0.055	−11.5	—
Terms	lead	+0.012	+2.0	—

The grid is close to a mirror: whichever axis carries the small signal under one way of counting the text carries the opposite sign under the other. That is worth stating plainly because the two halves are easy to mistake for each other — the theme-scored *lead* coefficient (−0.058) and the term-scored *atypicality* coefficient (−0.055) are nearly the same number attached to different quantities.

Nothing here reaches a tenth of a standard deviation. Taking the largest, a firm moving from no patents to twenty shifts its mean theme-scored lead by about 0.18 standard deviations, and the raw correlation across firm-years is −0.020. The weighting of the reference window is irrelevant to all of it: scoring against an exponentially decayed reference instead of a flat one moves every coefficient by less than 0.004.

One objection is timing: a patent is applied for before its invention is in public use and a trademark when the firm is ready to sell, so patents would lead. Firms do not proceed that linearly, and in some industries the trademark comes first (Seip et al., 2018). Entering patents at $t-2$, $t-1$, t and $t+1$ — the last allowing the trademark to precede the patent — one offset at a time, since the count is strongly autocorrelated within firm, gives −0.054 to −0.089 standard deviations of log patenting under topic scoring and +0.012 to +0.024 under term scoring. No ordering isolates a direction: the offsets are flat within each representation and opposite in sign between them. The largest coefficient anywhere is under a tenth of a standard deviation.

The second objection is heterogeneity: a pooled near-zero is consistent with genuine orthogonality everywhere, but equally with a positive slope in sectors where the two rights are complements — pharmaceuticals, chemicals, medical devices, machinery — offsetting a zero or negative slope where trademarks do the work alone. Classes are sorted in advance into complements, substitutes and neither, by judgement from the class definitions rather than by anything estimated: the eleven covering chemicals, pharmaceuticals, machinery, electronics, medical apparatus and scientific services are complements, the consumer-goods and services classes substitutes, the rest neither. Re-estimating the identical specification inside each group gives −0.000 ($t = -0.03$) among complements, −0.026 ($t = -1.45$) among substitutes, and an interaction of −0.008 ($t = -0.49$). Across the 27 Nice classes clearing a 50-owner floor, the complement classes run from sixth place to twenty-fourth by coefficient and do not agree in sign with one another. Within that 27-class set the only cells that reach significance are paper and housewares, both negative and neither a complement class. On the wider 34-class retention used for the sectoral build, the significant positives are transport and cosmetics — again neither a complement — while medical devices and machinery turn up among the significant negatives. Neither retention supports the complementarity story.

The test is conditional on a successful assignee match, so every owner patents at least once and the complement and substitute cells resemble each other more than the industries they stand for do, pushing any interaction toward zero. The weight rests on the cell-level nulls — among the firms where a patent–vocabulary relationship should be easiest to find.

4.7 Each axis bites at a different stage

Table 7 re-estimates every outcome on one harmonized sample — one row per debut owner, scored on that owner's first filing — under a single specification, so the coefficients can be read down a column. Magnitudes differ from the section-specific tables above.

Table 7: Every outcome in the funnel, conditioned on its prerequisite, under one specification. Unit is the debut owner, scored on that owner's first filing. Each row is a linear probability model with class×debut-year fixed effects and heteroskedasticity-robust errors; coefficients are percentage points per standard deviation. *Atypicality* is the average of the two KL levels; *lead* is the signed difference. Cohort windows differ because each maintenance gate needs elapsed time and Form D is electronic only from 2009, so the rows are not a nested decomposition. The first gate is a dated cancellation for non-use; the second is whether a registration that cleared the first was renewed rather than cancelled or expired, restricted to 2002–2013 cohorts, since a year-six and a year-ten death share a terminal status and are separable only this way. This table alone is estimated at $T = 200$ topics rather than the $T = 50$ used elsewhere; the sign and ordering of every row are unchanged at $T = 50$.

			Unsigned		Signed
Outcome *(population)*	n	Base	Atyp.	$\lvert\Delta\mathrm{KL}\rvert$	Lead
Reached registration *(filed)*	3,161,552	56.11%	−1.352 (0.033)	−0.307 (0.044)	+0.872 (0.028)
Passed gate 1, yr 6 *(registered)*	1,260,392	41.82%	+1.349 (0.054)	−1.150 (0.082)	−0.487 (0.052)
Passed gate 2, yr 10 *(gate 1)*	297,944	55.92%	+1.296 (0.108)	−0.297 (0.162)	−0.629 (0.105)
Raised Reg D round *(filed)*	1,535,004	2.53%	+0.003 (0.015)	+0.148 (0.027)	−0.149 (0.016)
Raised Reg D round *(registered)*	916,448	2.61%	−0.017 (0.020)	+0.115 (0.035)	−0.114 (0.021)
Listed after round *(funded)*	38,886	3.49%	+0.493 (0.129)	+0.608 (0.170)	−0.513 (0.104)
Became SEC-reporting *(registered)*	1,773,938	0.56%	+0.042 (0.007)	+0.040 (0.011)	−0.004 (0.007)

Standard errors in parentheses, in place of significance stars; see the note to Table 3.

Surprising language costs at registration and pays afterwards: atypical debut filings complete registration less often (−1.35pp per standard deviation), then survive the first use-proof gate more often (+1.35pp) and reach public markets more often (+0.04pp on a 0.56% base — the one specification in which a positive atypicality coefficient on listing survives, and it is a continuous term through a non-monotone profile, not the quintile contrast of Table 6). The examination process penalizes surprise; the market does not. Having been early runs the other way: leading applications complete registration *more* often (+0.87pp) and then pass the use-proof gate *less* often (−0.49pp on passing). Both survival rows point the same way at the year-ten renewal, four years later and on a sixth of the sample: atypicality +1.30pp and lead −0.63pp among marks that had already cleared the first gate. Whatever the first gate is selecting on, the second gate selects on it again rather than reversing it. Those filers follow through on the paperwork; what does not follow through is the product.

A linear coefficient cannot represent the inverse-U documented in Section 4, where both vocabulary tails complete registration less often than the middle. The positive linear term reflects the asymmetry of that shape — the leading tail completes more often than the lagging tail — and not a monotone gain from leading.

In the financing rows atypicality does nothing at one stage and a great deal at the next. Whether a debut owner ever raises a Regulation D round — a private securities placement, reported to the SEC on Form D — is unrelated to how atypical its filing was: −0.017pp per standard deviation among owners who completed registration, against a 2.61% base, which is indistinguishable from zero ($t = -0.8$). Among owners who did raise one, the same measure predicts *reaching* the public market: +0.49pp on a 3.49% base, about 14% relative ($t = 3.8$). Atypical language is invisible at the financing stage and informative afterwards, which is what an investor screen that does not read the text would look like — the text being free and public at the moment of filing. The comparison is among survivors, so it cannot separate that reading from selection on unobservables; and the null at entry is a null, not evidence of indifference.[6]

That last row turns on one construction. An exchange-registration marker on its own does not mean a firm listed *after* raising: 56% of matched owners carrying one registered before their first Regulation D notice, because established public companies also place securities privately. The row is therefore restricted to owners whose exchange registration postdates their first notice, which halves the coefficient and leaves it significant.

Three limits bound the financing rows. Regulation D notices are electronically filed only from 2009, so they rest on 2009–2018 debut cohorts. Owners are matched by normalized name, which succeeds for a small and unrepresentative minority favouring formally constituted entities, so every estimate is conditional on matchability. And the final row conditions on funding, realized after the filing being scored, so the coefficient is a contrast among survivors rather than a treatment effect: it establishes that information distinguishing eventual listers was already present in the text years earlier, not that the language caused the outcome.

Lead, by contrast, does not reverse anywhere after registration. It is negative at both use-proof gates, at financing with or without registration imposed, and at listing among the funded; the only row where it is not negative is SEC reporting among registered owners, where it is indistinguishable from zero (−0.004pp, SE 0.007). Being early is penalized at every stage where the market, rather than the examiner, is doing the selecting, and never rewarded at any of them.

5 Business vocabulary flows out of software, and entrants do not carry it disproportionately

The spatial version of this question is answered: von Graevenitz et al. (2022) show that novel description tokens reach distant US locations more slowly than near ones, in the *Regional Studies* programme surveyed by Castaldi & Mendonça (2022). The cross-industry version is open, and the Nice class system supplies the partition.

For nine curated business-model phrases, Table 8 records the first year each of four classes (software 009, tech services 042, transport 039, advertising/retail 035; 2.32 million granted filings) crosses a 1% prevalence floor. Software- and tech-services-origin phrases arrive in the physical-economy classes with lags of one to thirteen years; the only counterexample in the set is on-demand, which originates in transport. The 1% floor is sensitive to class size, and the asymmetry is a four-class result a 45-class build would test properly.

Unsupervised theme extraction corroborates the pattern without depending on which phrases were picked. A detection-purpose latent Dirichlet allocation (LDA) topic model ($T = 50$ on a stratified 200,000-filing sample) yields recognizable business-model components; for each (theme, class) cell crossing the 1% floor with six or more tracked years, Bass-model fits (Bass, 1969) on 118 retained cells give median innovation coefficient $p = 0.018$, median imitation coefficient $q = 0.110$, and median $q/p \approx 6.5$, in the range typically reported for consumer-product adoption. Tagging each multi-class theme's origin class and counting directed origin-to-arrival edges: software contributes

[6] This pair is sensitive to the reference window. Under annual reference buckets the entry row was −0.06pp ($t = -2.8$), which read as active negative selection. Per-filing windows move it to zero and leave the listing row intact, so the negative selection was an artifact and the informativeness among the funded was not.

Table 8: First year a theme reaches 1% prevalence among granted filings, by class. "–" = never clears the floor.

Theme	software	tech-svcs	transport	advert/retail
as-a-service	2011	**2009**	2013	2012
mobile-app	2010	**2009**	2012	2012
cloud	2012	**2010**	2011	2014
streaming	**2008**	2008	2013	2014
artificial-intelligence	2017	**2016**	2018	2018
augmented/virtual-reality	**2015**	2016	2022	2022
blockchain / crypto	2021	**2018**	–	2022
real-time	**2001**	2003	2009	2014
on-demand	2017	2016	**2014**	2017

33 of 72 edges against 18 expected under symmetry, and receives only 10. The asymmetry reproduces under independently-fitted non-negative matrix factorization (NMF) on the same sample; both are bag-of-words factorizations, so the robustness is within-paradigm.

Entrants carry the canonically new themes, but no more often than they carry anything else. The 50 earliest carriers per theme are 73.7% entrants on average — but the corpus's year-matched debut baseline is 76.4%, so the theme-carrier excess is −2.7pp and 28 of 50 themes sit *below* baseline. The themes leaning incumbent include environmental/sustainability, natural gas and petroleum, artificial intelligence, and cloud computing. So the arrival of new vocabulary is not an entrant phenomenon in the way the Schumpeterian reading would predict (Schumpeter, 1942).

The AI era has so far compounded rather than erupted, though the most recent data test that reading rather than confirming it. Dating each era from the first year its terms appear in at least 0.4% of filings in the two classes — 1994 for the internet, 2015 for AI, in both cases the year before the share at least doubled — the two track closely through year three, then diverge: internet vocabulary tripled between years four and six to 23% of filings in 2000, while AI vocabulary reaches 12.9% at year nine, roughly where the internet stood at year five (Figure 7). The last two observed years are the steepest in the series — prevalence rises about 14% a year to 2021, then 50% between 2022 and 2023 and 26% into 2024, following the release of consumer chatbots. That is an acceleration rather than a discontinuity, since per-year corpus turbulence shows no AI-era deviation through 2024, but it runs in the direction the eruption reading predicts. The comparison is sensitive to the era-start alignment, and the falsifiable marker is close to binding: if AI follows the internet's path with a lag, its prevalence should triple within roughly three years of the 2024 edge and the entrant share of its carriers should rise above baseline.

6 Lead tracks commercial risk; atypicality does not

Vocabulary position carries two separable signals, and it is the signed one that tracks what becomes of a product. Lead marks commercial risk: +1.4pp top-versus-bottom quintile at the use-proof gate with fixed effects, controls, and owner clustering, and −0.047pp per standard deviation in SEC reporting at equal atypicality, so the filings whose language an industry went on to adopt belong disproportionately to products that do not survive and to owners who do not reach public reporting. Atypicality runs the other way and on a different margin: it marks marks that survive their gate (+1.3pp per standard deviation), and its quintile contrast on SEC reporting is zero to three decimals once class, year and length are absorbed (Table 6). A continuous per-standard-deviation term on the harmonized sample of Table 7 does return +0.042pp on a 0.56% base; that is the one specification in which it survives, and it is a linear term through a non-monotone profile.

ΔKL, adapted from Barron et al. (2018); Murdock et al. (2017) onto topic distributions of USPTO trademark filings, produces interpretable structure on commercially-incentivised legal text. The construct is empirically distinct from patent-track invention within firm: no stable association survives a change of text representation, and no estimate reaches a tenth of a standard deviation. And the corpus supports direct measurement of cross-industry vocabulary diffusion, with multi-year origin-to-arrival lags, Bass-fittable adoption, and asymmetric flow out of software on the four classes built so far.

ΔKL does not measure innovation in any strong sense; the defended primitive is vocabulary position on commercial text, and Appendix G shows concretely which kinds of novelty the primary scoring cannot see. ΔKL does not identify an exploitable extreme-tail return premium; the large debut-specification figure fails a sample-size diagnostic. And the term-scored signed firm-level advantages of leading filings (margins, listing) are artifacts of drafting style and lexical specificity (Appendix C).

6.1 Five neighbouring constructs this is not

Novelty and *transience*, in the sense of Murdock et al. (2017) and Barron et al. (2018), are the two KL levels themselves: novelty is divergence from the past, transience divergence from the future. Atypicality as used here is their average, and so is neither of them alone; *novelty* in particular is declined because it names only the backward-looking half of a quantity that faces both ways. *Resonance* is their signed difference, and is arithmetically what this paper calls lead. The word is avoided because it names a mechanism — the field resonating with an author's contribution — that this design cannot observe. A filing can sit where its class was heading because it was imitated, because it and the class responded to the same outside event, or by coincidence; the measure does not separate these, and *resonance* would assert the first.

Innovation is a property of the product, not of the sentence describing it: two firms selling the same thing can draft very differently, and a genuinely new product can be described in boilerplate (Appendix G). The measure sees prose.

Radicalness (Dewar & Dutton, 1986; Henderson & Clark, 1990; Tushman & Anderson, 1986) is the closest neighbour, since KL is a departure measure rather than a newness count and the Amazon exhibit is a large departure built from unremarkable components. It is declined because it claims that existing competences are devalued, and nothing here licenses the step from unusual language to devalued competence.

Distinctiveness has two prior occupants. In trademark law it is the registrability doctrine, proven in part by the continued use this paper uses as an outcome. In strategic management, optimal distinctiveness (Zhao et al., 2017) predicts an inverted-U in conformity — a shape that appears here at registration but not at listing, so the term would imply a theory is being tested when it is not.

What *atypicality* says is narrower than any of them: this description is unusual for its class and year. Whether the thing described was new, radical, or good is not measured.

6.2 Filings are not independent draws

Some of them share an exact position with another filing, and outcomes cluster within owner — the within-owner correlation is 0.42 for gate survival and is mechanically 1.0 for public listing, since listing is a property of the owner and a firm with hundreds of marks contributes hundreds of identical successes. The estimates above are unaffected, because the gate regressions cluster on normalized owner and the firm-level specifications carry one observation per owner. But binning filings without that correction overstates how much structure the corpus contains: on a positional grid, correcting cell errors for the clustering design effect — the factor by which within-owner correlation inflates the variance of a cell mean — cuts apparent excess dispersion by roughly a quarter on the gate outcome and by half on listing. Displays of the vocabulary plane are descriptive here for that reason, and the near-zero pooled correlation between A and L reported in Section 3 does not survive cell by cell within class, so per-cell contrasts should not be read as within-class effects.

6.3 What remains unresolved

What drives the episodic pattern is not settled. Cohort and gate year move together, so post-crisis bet composition, app-era speculative filings, and calendar-time enforcement changes cannot be separated on this design. The representation is also fit on the pooled corpus; a vintage refit would close the remaining look-ahead channel.

7 Reproducibility

Code, firm-year panels, and analysis outputs are at `https://github.com/ericsilver/tm-vocabulary`, indexed in the repository `README.md`. The pipeline rebuilds from a USPTO Open Data Portal API key with a single `make tm` command (the TRTYRAP backfile is about 12 GB, streamed once). Published panels include the SEC-matched firm-year ΔKL panel ($n = 59{,}033$) and the PatentsView-joined panel ($n = 174{,}569$). An online appendix under `docs/online-appendix` carries a three-panel breakout for each of the 45 Nice classes, the cross-industry comparisons, and a machine-readable table of per-class contrasts with standard errors.

A Computation, corpus, and robustness

A.1 Corpus and scoring

The USPTO Trademark Annual Applications backfile (product TRTYRAP) provides a complete record of US trademark filings from 1884 onward. The deepest diagnostic work uses the complete software/electronics class (Nice 009; 1.81M filings); the diffusion application uses four classes totalling 2.32 million granted filings 1990–2024; the gate analyses use all 45 classes. The firm-level panels link trademark owners to SEC EDGAR financial filers and to PatentsView patent assignees by normalized-name join.

The topic representation is a $T = 50$ LDA fit on a stratified sample of 448,437 filings across all 45 classes (vocabulary of 62,168 unigrams and bigrams); every filing 1990–2024 is transformed to a dense 50-topic distribution, and past-facing and future-facing surprise are computed against per-filing topic references spanning 1,826 days on each side of the filing date, scoring 1995–2019. A filing whose window holds fewer than 500 same-class filings on either side is left unscored, since a thin reference inflates the divergence mechanically; that floor, together with the corpus edges, is why roughly two-thirds of filings carry a score. A term-level variant is retained alongside, on the same anchoring: token distributions (within-class document frequency ≥ 50) against per-filing windows of the same 1,826 days, Dirichlet-smoothed ($\alpha = 1$) over the class vocabulary. Because a goods/services description runs to a few dozen words, the divergence is needed only at the terms a filing actually uses, so the reference is carried as a running count vector and read sparsely. The term variant is used where vocabulary identity is the object (phrase transit) and as the comparison scoring throughout. Both representations are also computed under an exponentially decayed reference with a two-year half-life, truncated at the same window; Appendix A.4 reports what changes.

A.2 Scores are interpretable from 1995

Burn-in is the span at each end of the corpus where the reference windows cannot be filled, so scores there reflect a thin reference rather than the text being scored.

The signal depends on having enough past and future filings to populate the references; the 5-year past window is incomplete before 1990 and the future window after 2020, and both zones are excluded (Figure 8). The burn-in cut is set empirically: thin past references inflate past-facing surprise, and profiling that bias across all 44 sufficient-volume classes gives a median absolute bias of $\sim$2.0 nats — the natural-log unit in which these divergences are measured — in 1990, 0.25 in 1991, 0.10 in 1992, and 0.04 from 1993 onward, below any interpreted effect size. The 1995–1997 deviations ($\sim$0.15–0.17) are not burn-in — mechanical contamination cannot rebound — but the dot-com era moving the corpus. Per-class burn-in lengths estimated under a stability-band criterion track each class's noise floor rather than its reference density, so a single standard cut at 1995 is used, conservative by about two years.

A.3 Lead is a small residual, and term scoring tracks length

Past-facing and future-facing KL are highly correlated however scored. On the complete class-009 corpus under the production scoring they correlate at $r = +0.988$ (n = 1,109,643); under term scoring on class-year references the figure is +0.971. The lead L is therefore a small residual of two large, stable quantities: both levels have a standard deviation near 0.70 nats about a mean of 1.46, while L has a standard deviation of 0.109 — about a sixth of either level's spread. Only 1.2% of the two levels' combined variance survives into their difference. That is what makes L sensitive to anything that inflates the levels, and under term scoring the dominant such thing is document length. Term-scored atypicality correlates with log filing length at -0.646 in class 009 and -0.587 in class 035, for the reason set out in Section efsec:construct: the divergence charges a filing for the narrowness of its own distribution, and that narrowness is fixed by how many distinct terms it uses. Theme scoring breaks the coupling, taking the same correlations to -0.084 and $+0.058$ — the second of them positive. The consequence for the lead is direct: the standard deviation of L rises threefold across deciles of A under term scoring and by a quarter to a half under topic scoring, so a unit of lead means roughly the same thing everywhere only in the topic representation. Appendix F shows this as a length surface over the two axes. Two qualifications. Topic scoring returns a finite value for about 65% of filings, so the two columns are not drawn from the same set; re-estimating the term-scored correlations on exactly the topic-scored subset barely moves them, which puts the contrast down to the representation rather than the sample. On that common subset the spread comparison is nearer twofold than the threefold quoted above. And "flat" means no monotone length gradient rather than constant: class 035 under topic scoring is mildly U-shaped in A. The two scorings agree only moderately per filing (Pearson 0.46, Spearman 0.36 on 1.9M jointly scored observations), which gives their agreement on the mark-level patterns evidential force and their disagreement on the firm-level patterns diagnostic force (Appendix C). Across $W \in \{3, 5, 7\}$, 6.9–10.4% of filings flip sign. Per-filing noise attenuates at every aggregation used: firm-year means, token pools, theme prevalence trajectories.

Alignment with expert innovation judgment is at most suggestive. Firms on BCG/MIT "most innovative" lists sit +0.09 standard deviations above the panel mean under term scoring ($p = 0.22$), +0.14 under topic scoring ($p = 0.07$), and +0.08 at $T = 200$ ($p = 0.20$). The matched sample is 41 firms, which is too small to carry weight in either direction, and the paper draws no inference from it. Patent counts remain uncorrelated with ΔKL on the same panel (Pearson +0.010).

A.4 The mark-level results hold under every scoring

The snapshot cross-check on the single one-gate-elapsed cohort (registrations 2016–2018, terminal status 710 versus 701–705/800) agrees with the event-dated estimates and is scoring-robust: −5.4pp survival across quintiles under topic scoring and −5.7pp under term scoring on the identical sample (Figure 9). Under term scoring, gate survival *rises* in the raw KL level (+6.5pp past-facing, +7.6pp future-facing): atypical text survives the use requirement better while leading text survives it worse — the same two-axis separation as Section 4.5, at the mark level.

A.5 Annual reference buckets overstate the penalty by a third

Every estimate uses a reference anchored on the filing's own date. The alternative is one reference per class-year, under which a filing made on 1 January and one made on 31 December receive identical references and neither sees any of the language filed during its own year. Appendix B reports the calendar signature that leaves.

First, the gate penalty was overstated by about a third. On the same 3.10 million registrations, the top-versus-bottom contrast is +2.15pp (SE 0.30) under annual buckets and +1.39pp (SE 0.17) under per-filing windows. The attenuation is remarkably uniform across specifications — the ratio runs 0.59 to 0.73 for every sample split in Table 3 — and it is an attenuation of magnitude only: standard errors fall by roughly half, so the t-statistic *rises*, from 7.2 to 8.3. The same holds in the software class alone, where the contrast moves from +7.40pp to +5.25pp.

Second, the shape of the gate penalty changes. Under annual buckets the quintile profile looked like a gradient with a steeper final step; under per-filing windows the gradient through the interior quintiles is shallower and the final step dominates, with the top quintile carrying about seventy percent of the total. The coarser measure spread a top-quintile effect across the whole range.

Third, one published shape does not survive. Registration completion showed an inverse-U on both the unsigned and the signed axis. Only the unsigned one is real; on ΔKL the curvature disappears entirely under per-filing windows, and the earlier interior peak is attributable to the month gradient above. The firm-level results are the least affected by the window change, though they are sensitive to the owner-to-registrant linkage instead (Section 4.5): the signed decomposition keeps both its signs and its significance under either window.

A.6 Signal is flat between five and seven years

The window is a parameter of the measure, not a property of it, and the five years used throughout was inherited from the class-year construction rather than chosen under the current one. Re-scoring the software class at several widths under per-filing references, and judging each by the first-gate quintile contrast it produces, gives +2.02pp at two years, +2.20 at three, +3.23 at five and +3.37 at seven, each on an identical sample of 452,911 registrations and each with a standard error of 0.23pp. Signal rises steeply from two to five years and is flat between five and seven: the +1.03pp gain from three to five is well outside the sampling error, the +0.14pp from five to seven is not. A ten-year window returns +2.89pp, but that figure is not comparable to the others — requiring both windows to lie inside the corpus costs 22% of the scored filings and 13% of the gate sample, dropping the earliest and latest registration years, which are also the years whose contrasts differ most. The apparent decline at ten cannot be separated from that change in composition and is not evidence about window width. On the range where the sample is fixed, seven years is nominally the peak and five is within noise of it while retaining more of the corpus, so five is defensible rather than optimal. The sweep is on one class and one outcome.

A.7 The gate punishes breaking with the past more than resembling the future

"Leading" is relative to a reference, and the two references need not be the same width. A short past and a long future make lead a measure of foresight — whether the filing resembles where the class was heading over the next seven years more than what it wrote in the last three. A long past and a short future make it a measure of rupture — whether the filing has broken with what the class had established over seven years, judged against only the next three. Scoring every class on the production representation and per-filing windows at $W \in \{3, 5, 7\}$ years on each side, and evaluating all five variants on the identical 3.10 million registrations with quintiles assigned within class×registration-year:

Variant	Past, future (years)	First-gate failure Q5−Q1 (pp)	First-gate failure classes > 0	Registration Q5−Q1 (pp)	Registration Q3 − tails
Foresight	3, 7	+1.65 (0.09)	31 of 44	+1.09	+0.50
Symmetric	3, 3	+1.69 (0.09)	—	+0.79	+0.13
Symmetric	5, 5	+2.29 (0.09)	33 of 44	+0.71	+0.28
Symmetric	7, 7	+2.58 (0.09)	—	+0.48	+0.65
Past-rupture	7, 3	+3.00 (0.09)	37 of 44	+0.01	+0.33

The symmetric five-year row is the paper's FE-only estimate in Table 3, reproduced here to the decimal, which fixes the scale. The penalty rises monotonically along the mix, from +1.65pp under foresight to +3.00pp under rupture, and rupture is the more pervasive variant, positive in 37 of 44 classes against 31. In software the two are +1.82 and +5.07pp. So the use requirement punishes distance from the class's established language more than it punishes resemblance to its coming language; but foresight carries a penalty of its own, more than half the symmetric one, and the separation is a matter of degree rather than kind.

Registration is a different matter. On the signed axis the top-to-bottom contrast is at most a point on a 57% base and the interior-minus-tails depth at most two thirds of a point, under every mix. An earlier version of this decomposition, run on the retired term-level calendar-year scoring, found a registration inverse-U of four to five points that the window mix did not touch; under per-filing references that curvature is gone on the signed axis, consistent with Appendix A.5, which traces it to the calendar artefact. The examiner's response to lead, on this scoring, is close to nil.

B Representation and resolution drive the score; weighting does not

B.1 Representation, anchoring, weighting: which of the three matters

A scoring is fixed by two choices. The *representation* is what the language is turned into: a distribution over 50 fitted topics, or a distribution over the class vocabulary's terms. The *reference* is what that is compared against, and it hides two sub-choices — the *anchoring* (one reference per class-year, or one per filing built on its own date) and the *weighting* inside the window (every day equal, or decaying by half every two years). Six builds exist:

Representation	Anchoring	Weighting	sd(lead)	sd(atypicality)
Topics	class-year	flat	0.132	0.672
Topics	per-filing	flat	0.102	0.698
Topics	per-filing	half-life 2y	0.082	0.698
Terms	class-year	half-life 2y	0.299	1.330
Terms	per-filing	flat	0.155	1.101
Terms	per-filing	half-life 2y	0.130	1.102

All six are compared on the 6.01 million filings every one of them scored. The second row is the production scoring. The fourth is the legacy build, and it is the one to handle carefully: it moves anchoring *and* weighting at once, so a result computed on it cannot be attributed to either. That is why the two per-filing term builds were constructed.

Changing one choice at a time settles the ordering. Correlations of lead between builds differing in exactly one respect:

Choice varied	r (lead)	r (atypicality)
Weighting, flat vs half-life 2y (topics)	0.982	1.000
Weighting, flat vs half-life 2y (terms)	0.970	1.000
Anchoring, class-year vs per-filing (topics)	0.738	0.929
Resolution, 50 vs 200 themes	0.636	0.828
Representation, topics vs terms	0.460	0.396

The weighting is close to irrelevant: whether recent language counts for more than distant language within the window barely moves the score, and it does not move atypicality at all to three decimals, which is what the 45-degree rotation of Section 3 implies — the long axis is a property of the filing and the weighting acts almost entirely on the small residual. Anchoring matters moderately. Resolution and representation matter most: quadrupling the number of themes leaves the two builds agreeing on 40% of the variance in lead, and topics against terms on less than a quarter. A robustness check that varies only the weighting is therefore close to a restatement; one that varies the representation or the resolution is a real test, which is why the term-scored replication in Appendix A.4 and the sweep below are the ones the claims lean on.

B.2 Fifty themes is coarse, and the finding does not depend on it

Fifty themes over a 62,168-term vocabulary is roughly one theme per industry, so the partition is coarse enough that a reader may reasonably suspect the findings belong to it. Raising the theme count does not add coverage — the vocabulary is fixed, and a larger T re-partitions the same words — but it changes what counts as sitting far from an industry, and could change any sign. Re-fitting the model at four resolutions on the same stratified sample and the same analyzer, and re-scoring the software class under per-filing windows:

Themes	Gate contrast, lead	t	Gate contrast, atypicality	r with $T = 50$
50	+3.23pp	13.8	−14.01pp	—
100	+2.71pp	11.6	−13.69pp	0.690
200	+5.20pp	22.2	−13.45pp	0.640
500	+5.69pp	24.3	−14.49pp	0.641

Three readings, in descending order of how much weight they carry. The sign and the significance are untouched across a tenfold change in resolution, and so is atypicality, which stays within a point of -14pp throughout. The magnitude is not: the lead contrast runs from +2.71 to +5.69pp and does not move monotonically in T, so no single resolution can be quoted as the software class's effect, and the figure this paper reports at $T = 50$ sits at the low end of that range rather than the high one. And agreement with $T = 50$ settles at 0.64 without decaying as T grows, which says the resolutions share a fixed core and differ in the remainder rather than drifting apart as the partition refines.

The corpus-wide contrast is steadier than the single class: at $T = 200$, scored for all 45 classes, it is +2.23pp against +2.29pp at $T = 50$. Only those two resolutions exist corpus-wide, so that is a two-point comparison, but it is the one the paper's estimates run on, and pooling evidently averages away much of the per-class resolution sensitivity.

B.3 Two fits at one resolution disagree almost as much as two resolutions

The sweep leaves one question open. Scores at different resolutions agree at $r = 0.64$ to 0.69 on lead, and neighbouring resolutions agree no better than distant ones. That has two readings. The number of themes may genuinely change what is being measured. Or a topic-model fit may simply be unstable — two runs settle in different local optima — in which case the sweep is measuring fitting noise and the resolution has little to do with it.

Holding T fixed and moving only the seed separates the two. A second fit at $T = 200$ on the same sample, the same vocabulary and the same number of passes, differing from the first in nothing but its random seed, agrees with it at $r = 0.79$ on lead and $r = 0.87$ on atypicality across 1.11 million scored software-class filings. The cross-resolution figures were 0.64 and 0.83. Most of what the sweep showed as resolution sensitivity is therefore present between two fits that differ only in the seed: for atypicality nearly all of it, for lead well over half. The resolutions are not measuring different things; each fit is a noisy draw on the same thing.

The estimate moves less than the scores do. On the identical gate sample, the first-gate lead contrast is +5.20pp under the first seed and +4.88pp under the second (SE 0.23 each), and the atypicality contrast -13.45 and -14.38pp. Two consequences follow. A single fit's lead score carries measurement error of roughly a fifth of its variance, which attenuates every coefficient on lead toward zero; the estimates in this paper are on the conservative side of what a noiseless score would return. And the spread of magnitudes across the resolution table above, which does not move monotonically in T, should be read as a band about a third of a point wide from fitting noise alone rather than as a property of any particular T.

Annual anchoring leaves a signature that per-filing anchoring does not. Mean lead by calendar month of filing, in units of the build's own standard deviation, has no reason to trend — nothing about filing in December rather than January makes a description more forward-looking:

Representation	Anchoring	December − January (sd)
Terms	class-year	+0.169
Topics	class-year	+0.041
Topics	per-filing	−0.008
Terms	per-filing	+0.017

A December filing sits eleven months further from its past reference and eleven months nearer its future one, and under annual buckets that shows up as position in the calendar. Under per-filing references it is gone.

The substantive claim does not depend on any of this. The first-gate contrast, computed on the common sample, runs +2.73, +2.71, +2.39, +4.05, +3.97 and +3.19pp on lead across the six builds in the order tabulated above, and −7.75 to −6.17pp on atypicality. Every sign holds; term scoring returns the larger lead contrast, so the headline estimate is the conservative representation.

B.4 One gate is linear; the public-reporting profile is a U

Four shapes were fitted to each profile by weighted least squares on 40 equal-count bins (Figure 10): linear, quadratic, an exponential $a + be^{cx}$, and a free-vertex symmetric form $a + be^{c|x-x_0|}$ meant to catch a U directly. Selection is by AIC.

Outcome	Axis	R^2 line	R^2 curve	ΔAIC	Shape selected
Registration	atypicality	0.38	0.92	77.1	curve, peak at +1.04
Registration	lead	0.08	0.17	1.9	(quadratic; weak)
Use-proof	atypicality	0.74	0.74	0.0	linear
Use-proof	lead	0.44	0.55	4.3	V, vertex −1.50
SEC reporting	atypicality	0.74	0.84	15.2	curve, peak at +2.17
SEC reporting	lead	0.13	0.59	26.3	V, vertex +0.30
listed only	lead	0.19	0.46	12.2	V, vertex +0.41
reporting, not listed	lead	0.07	0.67	37.2	V, vertex +0.24

The winning form is a free-vertex $a + b\,e^{c|x-x_0|}$. On the atypicality rows it is a genuine curve, with fitted exponents near 0.5 to 0.9 — a hump that peaks about one standard deviation above mean atypicality at registration and about two above it at SEC reporting. On every lead row the exponent collapses to the order of 10^{-4}, which turns $e^{c|x-x_0|}$ into $1 + c|x - x_0|$: what wins there is a *V*, linear in distance from a vertex. A V beating a parabola says the turn is sharper than a quadratic allows, not that anything is multiplying. The script now flags that degeneracy rather than leaving it to be read off the fitted constants.

Vertices are in standard deviations of the axis across filings. An earlier version of this table standardized on the spread of the forty bin centres instead, which is narrower than the filings' own spread and put the vertices on a scale that did not match the per-standard-deviation slopes quoted elsewhere in the paper.

The two rows for SEC reporting are the outcome Section 4.5 and Table 6 actually run on; the listed and reporting-only splits below them show the same shape on each half, which is why the split does not change the reading.

AIC here is computed on 40 binned means, not on the millions of underlying filings, so it ranks shapes rather than testing them. The 77 at registration and the 12 to 36 at the public-reporting gates are decisive; the 1.9 at registration on lead and the 4.3 at the use-proof gate are weak, and those two rows are better read as "no worse than linear" than as curvature.

B.5 Listing and merely reporting behave the same way

"Appears in SEC records" mixes two populations: companies whose shares trade, and entities that file with the Commission without ever listing — private issuers, funds, subsidiaries registering securities. Splitting the 19,889 matched owner strings on whether their CIK carries a ticker in the Commission's own company-ticker file gives 10,556 listed against 9,333 that file financial statements without one.

It does not change the finding. Among registered debut owners whose debut filing is scored, 0.310% later appear as listed companies and 0.285% as reporting non-listed ones, and both profiles are U-shaped in lead with vertices at +1.35 and +0.56 standard deviations. The listed profile is slightly flatter and noisier, which is what the smaller and more selected outcome predicts. Whatever the signed axis is picking up at this margin, it is not an artifact of counting non-operating registrants as commercial successes, and it is not specific to the act of listing either.[7]

C The measurement hazard: term scoring manufactures firm-level signal

Under term-level scoring, leading filings appear to belong to better firms. Among registered debut filings, the probability that the owner ever appears in SEC EDGAR rises in term-ΔKL from 0.29% at the bottom quintile to 0.42% at the top; on an SEC-matched firm-year panel, trailing 3-year mean term-ΔKL predicts gross margin at +2.3pp per standard deviation ($t = 2.7$ on the locally rebuilt panel through 2019; +3.2pp, $t = 5.3$, on the published panel through 2024), with past-facing surprise entering positively and future-facing negatively; and firm-year term-ΔKL predicts excess stock returns of +1.5 to +5.0pp per standard deviation at one-to-four-year horizons. A larger debut-only return figure (+28pp at 4 years) fails a sample-size diagnostic and is not claimed.

None of the signed versions survives distribution scoring. On identical samples, the debut listing gradient in signed topic-ΔKL is flat to declining, and the margin coefficient reverses: −1.3pp per standard deviation ($t = -1.8$) at $T = 50$, −2.2pp ($t = -3.1$) at $T = 200$, and −0.8pp ($t = -1.1$) at $T = 500$, against the term-level +2.3pp; the past-/future-facing decomposition flips sign and stays flipped. The term-level listing gradient separately dissolves under document-length controls: within length bands it is U-shaped rather than rising, with the monotone rise contributed by composition across lengths (listed firms' counsel write longer descriptions). The mark-level results are unaffected — they hold under both scorings at every resolution.

Vocabulary blindness does not rescue the signed signal. The topic model's vocabulary is sample-built, so rare and emergent terms are invisible to it; if the firm signal lived exactly there, topic scoring would erase a real signal rather than an artifact. Computing each filing's invisible-vocabulary share (63.8% of class-009 term *types* but only ∼9% of token *mass*): the share itself is a strong firm marker (P(EDGAR) 2.9% to 5.5% across its quintiles; gate survival +5.7pp), but *within* invisible-share strata the signed ΔKL gradient on the firm outcome is flat to negative, while the gate penalty holds in both strata (−6.9 and −8.3pp). The term-level firm signal is *unsigned specificity*: serious firms describe specific products with specific, therefore rare, words, and rare-term mass inflates term-ΔKL mechanically. Topic count is a partition dial, not a coverage dial — a crypto/blockchain topic exists at $T = 200$ but not at $T = 50$ or $T = 500$, because raising T re-partitions the fixed vocabulary along dominant axes — so resolution sweeps test granularity, not coverage, and the invisible-share split is the test that binds.

The registration filter does not explain the divergence. Dropping the registration conditioning leaves the term-level listing gradient essentially unchanged (0.26% to 0.38% across quintiles against 0.29% to 0.42% conditional), and an examiner channel touching 4.3% of failed applications cannot generate several-point quintile gaps.

Term-resolved text-novelty measures can therefore manufacture firm-performance correlations from drafting style and lexical specificity, and a distribution-based rescoring is a cheap, decisive diagnostic.

[7] Two limits on the split. The ticker file is a current snapshot, so a company that listed in 2003 and was acquired in 2011 appears in the non-listed tier; the tiers are therefore "currently listed" rather than "ever listed," which will misclassify older successes downward. And the base rates here are computed on the appendix's own panel, which conditions on the debut filing carrying a score and so is smaller than the body's sample.

D Examination rewards a middling specificity

Across 3.59 million debut filings (57.6% base rate), completion is an inverse-U in *atypicality*: it climbs from 46.6% at the least atypical end to about 58% in the middle and falls back to 53.4% at the most atypical (Figure 2, panel C). The same shape appears on the past-facing level alone, more mildly, running 54.0% at the bottom quintile to 58.4–59.1% in the middle and 57.7% at the top (Figure 1A).

The signed axis behaves differently. Completion on ΔKL falls to a trough of 51.9% just above the median and then rises steadily to 58.0% at the leading end. That trough is composition rather than timing: $|L|$ correlates with atypicality at +0.31, so filings near zero lead are disproportionately low-atypicality filings, which complete poorly for the reason above. Split by atypicality, the trough is absent in the upper three fifths and survives in the lowest. Reference windows built by calendar year put an inverse-U on this axis too, but that one is an artifact of annual bucketing (Appendix A.5) and does not survive at daily resolution. The signed axis still enters a linear specification strongly (Table 7), but monotonically rather than through an interior peak: what curvature exists at registration belongs to how far a filing's language sits from its category, not to which side of it the filing sat.

The obvious objection is that unfamiliar language is simply refused more often, so the measure recovers nothing beyond familiarity. Completion falls at *both* ends, so the most category-typical filings also complete less than the middle, which familiarity cannot explain; and the curvature is specific to the unsigned axis, the signed axis showing no interior peak at all.

Refiling after abandonment is itself mildly related to vocabulary position, though not on the axis that matters here. Refiling rates are flat in atypicality (a middle-minus-tails difference of −0.06pp on a 3.6% base) but rise with lead, from 2.6% among the most lagging abandoned applications to 3.4% among the most leading. Firms whose language ran ahead of their industry are slightly likelier to try the same mark again.

E The outcome analysis without registration conditioning

Figure 11 repeats the lifecycle outcome analysis without the registration conditioning, on filings 2013–2015 pooled across all classes (a window whose registrations land by ∼2017 and face at least one Section 8 gate by the 2025 snapshot; the earliest registrations in the window also face their year-ten renewal, so panel B mixes one and two gates). Panel A is the unconditional composite, the probability that a filing both registers and passes the gate, measured over all filings including the never-registered. It is nearly flat in ΔKL — 22.4% at the bottom quintile against 22.1% at the top, a middle-versus-tails depth of +0.001pp (SE 0.10) — because two opposed gradients cancel: registration rises in lead (61.3% to 63.2%) while gate survival falls (44.2% to 40.8%). Conditioning on grant is what separates them. Panel B shows the conditional gate outcome on the same filings, and it is U-shaped rather than inverse-U (depth −2.11pp), with a clearly penalised leading tail consistent with the cleaner single-gate cohort of Figure 9.

F Term scoring measures length; theme scoring does not

Figure 12 puts the two representations side by side over the same axes, with each hexagon coloured by the mean length of the filings in it. Under term scoring the colour runs from dark near the origin — a median of about 200 terms — to near-white at the far end, where the median is eight.

That gradient is the measurement problem: a filing scores as atypical largely because it is short. Under topic scoring the same surface is a narrow band of roughly uniform tone.

Two filings are marked in both columns. Under the production scoring they behave differently from one another, which is the point. AMAZON.COM (1997) is a leader on both representations — the 98th percentile of its class on terms and the 94th on themes — while sitting at the 40th and 47th percentiles of atypicality, so both agree that it was early and neither mistakes it for strange. Apple's IPOD (2001) divides them: the 31st percentile of lead on terms against the 62nd on themes. The surfaces plotted here are on the class-year build the figure was generated from, where the length gradient is starkest; the percentiles just quoted are on the per-filing build used for every estimate.

Topic scoring is independent of document length, which term scoring is not; term scoring resolves novel word *combinations*, which topic scoring does not. Hence topic scoring for every estimate, since the length confound contaminates firm-level comparisons, and term scoring for every worked example, since the topic basis has no coordinate for the language the examples turn on.

G How three filings encode

Table 9 traces three filings through term scoring and topic scoring at $T \in \{50, 200, 500\}$ — three kinds of novelty, and how each representation handles them.

AMAZON.COM (1997, advertising/retail) is *combinatorial* novelty: every word is ordinary, but the bigrams assemble an invention ("computerized line," "line search," "search ordering"; class document frequencies 379, 134, 355). The filing's text reads "on line" as two words, the standard spelling in 1997, before *online* lexicalized into a single token; the analyzer drops the stopword "on" and the surviving bigram "line search" is the act of forming the compound caught in progress. The novelty was necessarily combinatorial because the word for it did not yet exist. Term scoring sees these bigrams; under topic scoring at every resolution all of them are out of vocabulary and the filing reads as a media retailer that uses computers.

KOMBUCHA (1997, beer/soft drinks) is *lexical* novelty: the new thing arrives as new words (*kombucha* df 567, *saccharose* out of vocabulary even at the class level). The word *kombucha* is in the topic vocabulary but, with no kombucha-like topic, its mass dissolves into a generic beverages topic (≥ 0.60 weight at every T).

ETHEREUM (2018, software) is *wave* novelty: emergent vocabulary already diffusing (*blockchains*, *distributed computing*). It loads on generic software and services topics at every resolution, including $T = 200$, which does contain a cryptocurrency/blockchain topic — the eleven occurrences of *software* outvote the blockchain terms.

Table 9: Top topic loading for three filings, by representation. Term scoring resolves the novel content of all three; topic scoring resolves none of it at any tested resolution.

	AMAZON.COM (1997)	KOMBUCHA (1997)	ETHEREUM (2018)
novel content (term)	line search, search ordering	kombucha, saccharose	blockchains, distributed computing
$T = 50$	video, computer, music (0.40)	beverages, fruit, drinks (0.64)	software, online, downloadable (0.33)
$T = 200$	entertainment, video, games (0.29)	beverages, drinks, coffee (0.60)	field, services, development (0.37)
$T = 500$	computer, software, data (0.22)	beverages, drinks, fruit (0.61)	software, online, computer (0.49)

The pattern across the row is the measurement point in miniature: topic scoring cannot represent combinatorial novelty (Amazon), dilutes lexical novelty (Kombucha), and majority-votes away wave novelty (Ethereum) even when a fitting topic exists. That the mark-level outcome results are nearly identical under both scorings despite this blindness is what licenses topic scoring as a robustness instrument; that the firm-level results diverge is what exposes them as artifacts of the lexical specificity term scoring additionally captures (Appendix C).

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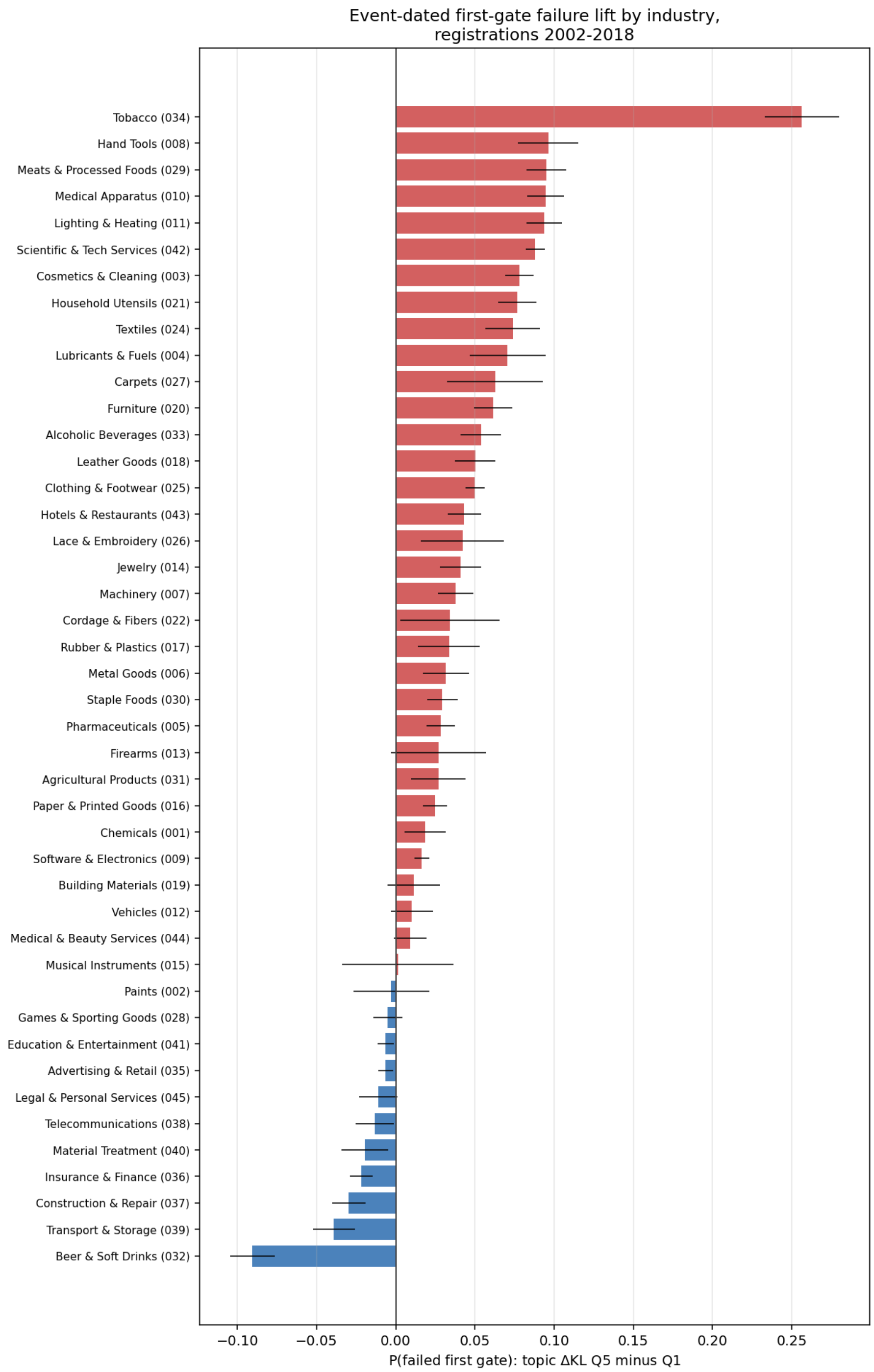

Figure 3: Event-dated first-gate failure lift (topic ΔKL top minus bottom quintile) by industry, registrations 2002–2018, raw quintile contrasts. Positive bars indicate the leading penalty. Whiskers are two-sample binomial intervals, uncorrected for the within-owner correlation (0.42) and so narrower than clustered intervals; read Table 3 for precision. Per-class breakouts are in the online appendix.

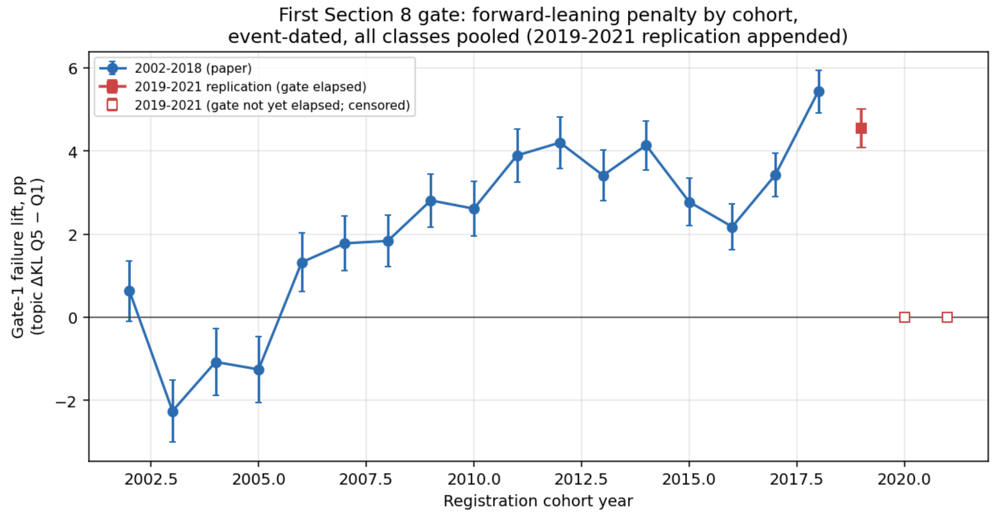

Figure 4: The leading gate penalty by registration cohort, all classes pooled. Each point is the event-dated first-gate failure rate of the most leading fifth of that cohort's registrations minus that of the most lagging fifth, with lead quintiles cut within the cohort and no controls. The penalty is absent or negative for the 2002–2005 cohorts, turns positive in 2006, and is present in every cohort from 2008 through 2019, ranging +1.8 to +5.6pp with no downward trend. 2019 is the last evaluable cohort: cancellations post at about age six and a half and the record ends in April 2026, so later cohorts have not reached their gate rather than having passed it.

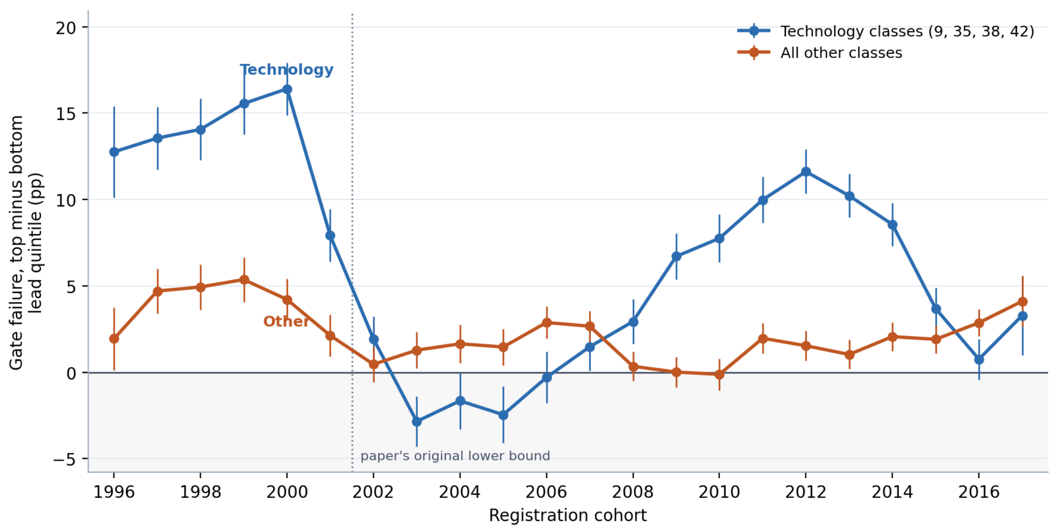

Figure 5: The leading gate penalty by registration cohort, technology classes (009, 035, 038, 042) against all others. Each point is the first-gate failure rate of the most leading fifth of that cohort's registrations minus that of the most lagging fifth, lead quintiles cut within cohort and group, no controls, with 95% intervals. Cohorts are restricted to registrations whose ninth anniversary is inside the record, so every outcome shown is settled. Outside technology the penalty is small and almost flat across two decades; inside technology it swings from +16pp for the 2000 cohort to −3pp for 2003 and back to +12pp for 2012. The dotted line marks the 2002 cut that the main cohort series uses.

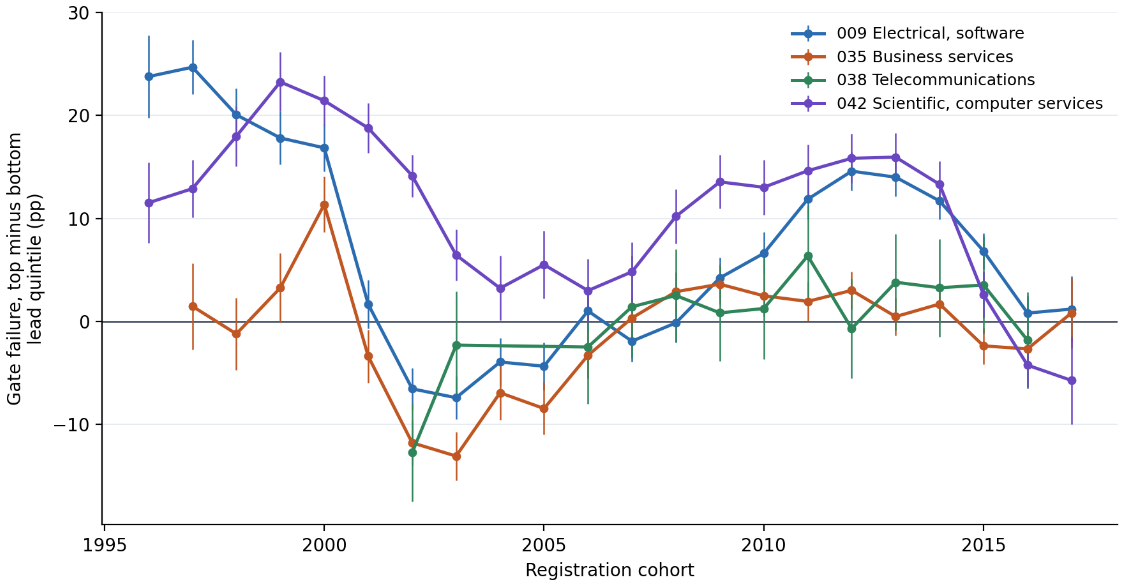

Figure 6: The leading gate penalty by registration cohort, one line per technology class. Each point is the first-gate failure rate of the most leading fifth of that class and cohort's registrations minus that of the most lagging fifth — lead quintiles cut within class and cohort, no controls — with 95% intervals, on registrations whose gate outcome is settled. Three classes cross zero between 2001 and 2006; scientific and computer services does not.

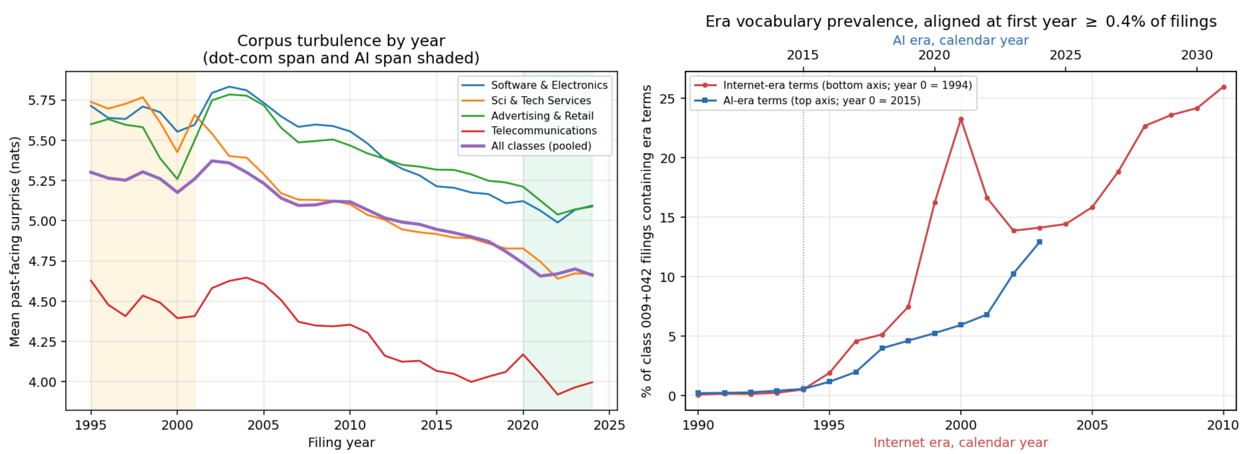

Figure 7: *Left:* per-year mean past-facing surprise by class and pooled, 1995–2024, with the dot-com and AI spans shaded. *Right:* share of class 009+042 filings whose goods/services text contains internet-era terms (internet, website, online, world wide web) or AI-era terms (artificial intelligence, machine learning, generative, neural network, chatbot, and kin), plotted in era years. Year 0 of each era is the first calendar year in which its terms reached 0.4% of the two classes' filings: 1994 for the internet, 2015 for AI. The two calendar axes share one scale, one year per unit of width; the AI series ends at 2024, nine years in, where the record ends, and has not yet run the internet's sixteen.

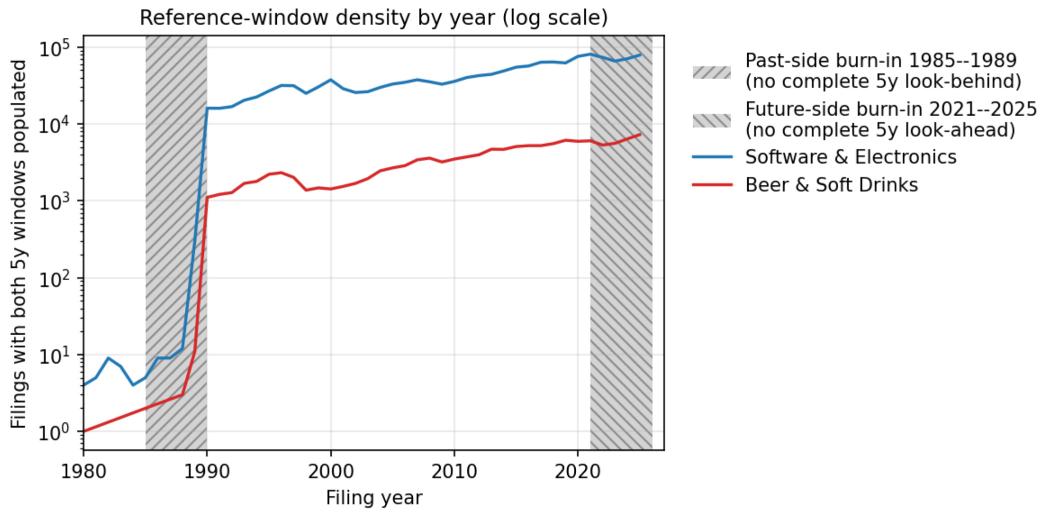

Figure 8: Reference-window density by year on log scale. Hatched regions are the past-side and future-side burn-in zones where the 5-year window is incomplete. The windows determine only how surprising a filing's language is — its score at the filing date. The outcome the score is tested against is read separately and later: whether the registration survived the use-proof gate, decided at registration age six to seven, which is why the score is interpretable from 1995 but the gate sample ends with the 2018 cohort.

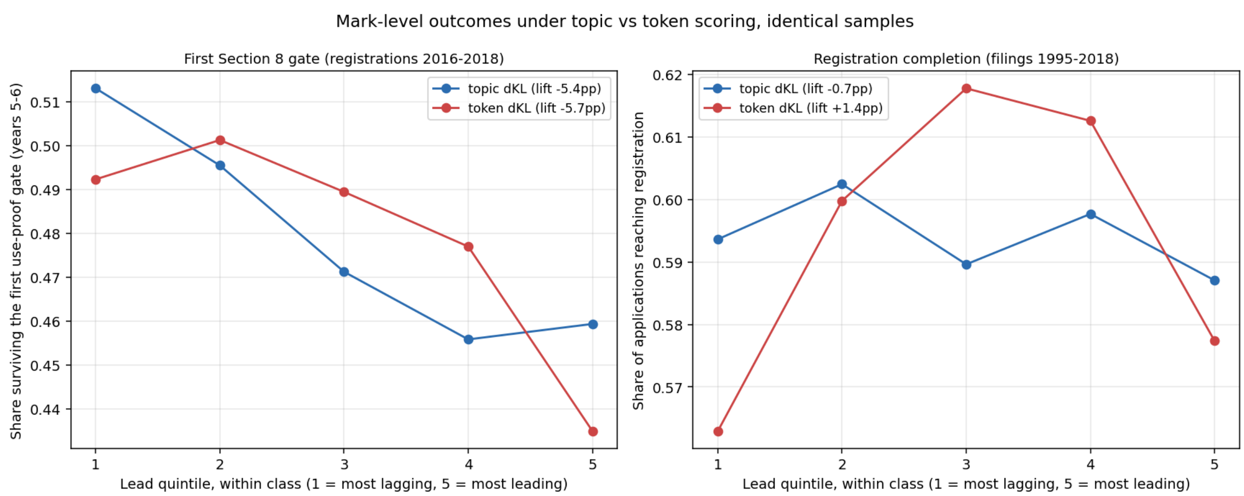

Figure 9: The two mark-level outcomes under topic and term scoring on identical samples, pooled across classes, by within-class lead quintile (1 most lagging, 5 most leading). *Left:* the share of registrations 2016–2018 that survived the first use-proof gate at years five to six, read from terminal status codes, where the two representations agree closely. *Right:* registration completion (filings 1995–2018), where they do not: the inverse-U in the signed axis is present under term scoring (depth $+0.048$) and absent under topic scoring (depth -0.001). The gate result is scoring-robust; the registration curvature on this axis is not, and Section 4 reports only the unsigned version.

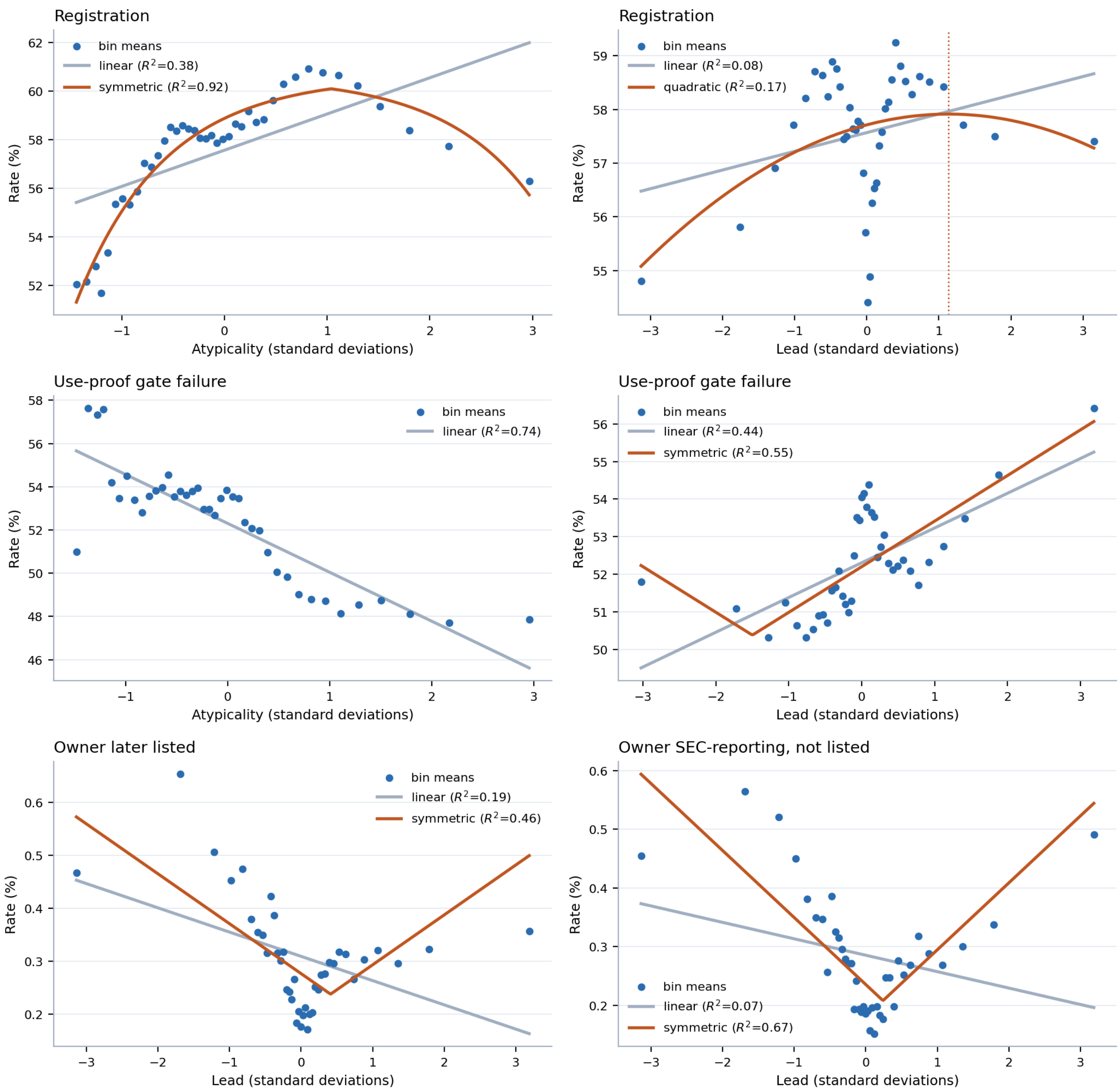

Figure 10: Binned profiles at the three gates, with the linear reading the paper's contrasts assume and the shape selected by AIC where that is not linear. Forty equal-count bins; the dotted line marks the fitted vertex. The bottom row is the case that matters: a linear fit through the public-reporting profile slopes downward while the data turn up at both tails.

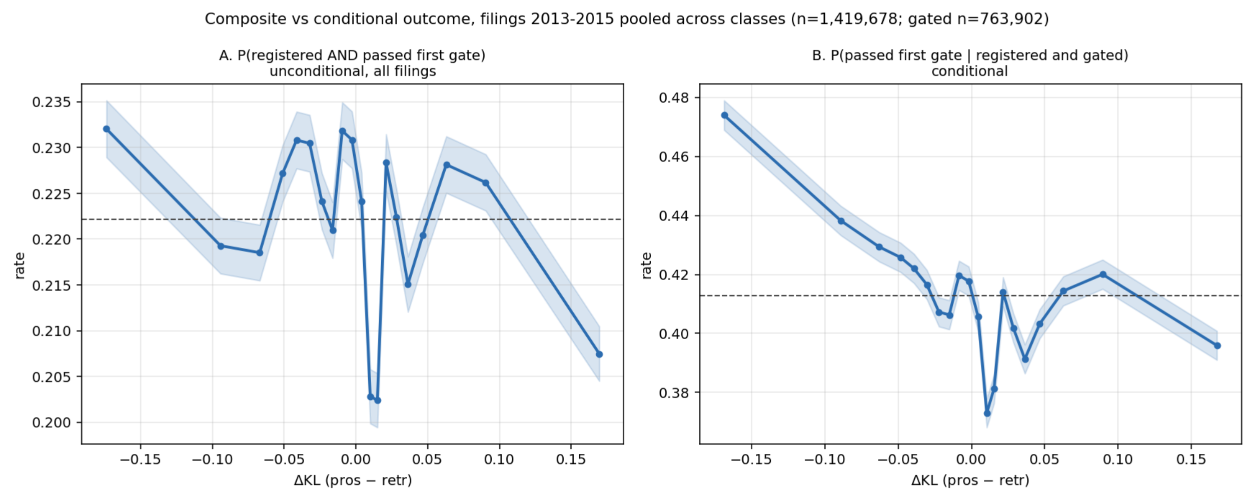

Figure 11: Filings 2013–2015 pooled across classes, topic-scored on per-filing reference windows (n = 1,419,678; gated subset n = 763,902). *A:* the unconditional composite P(registered and passed first gate) over all filings. *B:* P(passed | registered and gated) on the same filings. The unconditional view is nearly flat in lead (22.4% to 22.1% across quintiles); conditioning on grant is what exposes the gradient.

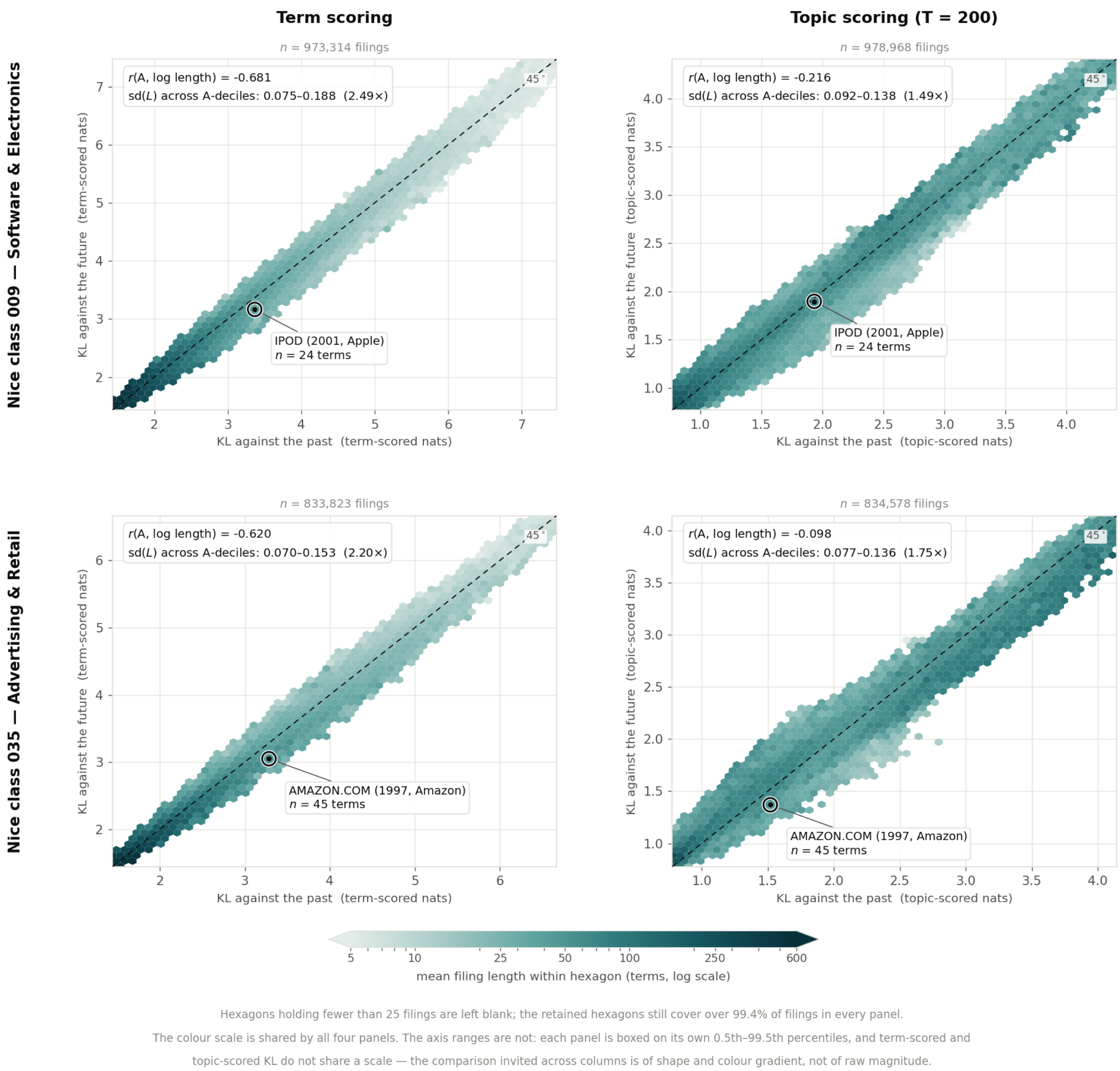

Figure 12: Mean filing length across the two axes, by representation and class, for filings 1995–2019 scored on per-filing reference windows. The horizontal axis is surprise against the preceding five years, the vertical axis surprise against the following five; a filing on the dashed diagonal is equally far from both, and distance below it is lead. Hexagons holding fewer than 25 filings are blank; the retained hexagons cover over 99.4% of filings in every panel. The colour scale is shared across panels; the axis ranges are not, since term-scored and topic-scored KL do not share a scale, so the comparison invited across columns is of shape and colour gradient rather than of magnitude. Both columns are drawn on the filings the per-filing windows score, which differ between representations by under one percent.